

Advances in nanotechnology in phase change materials and their applications

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ABSTRACT

The integration of nanotechnology into phase change materials (PCMs) has substantially advanced thermal energy storage and thermoregulation technologies across diverse sectors, including building construction, electronics cooling, smart textiles, and renewable energy systems. This review provides a comprehensive and critical synthesis of recent progress in nano-enhanced phase change materials (NePCMs), focusing on how nanomaterial incorporation modifies thermophysical behavior, durability, and application performance. Various nanomaterials—such as graphene, carbon nanotubes, and metal oxides are examined for their roles in enhancing thermal conductivity, regulating phase transition kinetics, and improving overall energy efficiency. Compared with conventional PCMs, NePCMs demonstrate markedly improved performance, with reported thermal conductivity enhancements of 20–160 % at nanoparticle loadings of 0.1–5 wt.%, while maintaining latent heat reductions within an acceptable range of <10–15 %. Hybrid nanomaterials and advanced encapsulation strategies, including in-situ polymerization and silica shell encapsulation, enable 10–30 % reductions in melting and solidification times and 30–70 % suppression of supercooling, alongside enhanced leakage resistance. Long-term reliability is evidenced by > 90–95 % latent heat retention after 200–500 thermal cycles. For thermoregulation applications, system-level energy efficiency improvements of 10–25 % are reported within operating temperature windows of 20–200 °C. Despite these advances, critical challenges persist, including nanoparticle agglomeration, increased viscosity, scalability constraints, material cost escalation of 10–40 %, and unresolved environmental and health concerns. This review also evaluates emerging applications in smart textiles, electronics, temperature-controlled packaging, and solar thermal systems, highlighting the cross-sector potential of NePCMs. Furthermore, it addresses sustainability considerations, emphasizing lifecycle assessment, green synthesis, and risk mitigation strategies. Novel insights into artificial intelligence-assisted property prediction and material optimization are discussed. Overall, this work provides a quantitatively benchmarked, application-driven roadmap for the sustainable development and industrial deployment of next-generation NePCMs.

1. Introduction

The manipulation of matter at the nanoscale, or between 1 and 100 nm, is the main emphasis of the multidisciplinary field of nanotechnology. By taking advantage of special qualities that materials display at this size, this technology has advanced a number of industries, including manufacturing, energy, and medicine [1,2]. Top-down, which involves carving bulk materials, and bottom-up, which builds substances

atom-by-atom, are the two main methods of nanomanufacturing [1,3]. Nanotechnology in medicine has particularly enabled significant advancements in medication delivery and imaging, addressing serious health issues such as cancer and cardiovascular disorders [2,4]. Rapid advancements in the creation of complex structures with nanometer accuracy hold promise for smarter goods and improved features in a variety of industries [4,5].

As compounds that absorb and release thermal energy during phase

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transitions, phase change materials (PCMs) are crucial for thermal energy storage applications in a variety of industries, such as solar energy systems, medicine, and construction. With the addition of nanoparticles, nanotechnology to PCMs improves their thermal properties, such as thermal conductivity, latent heat, and specific heat. This creates nano-composite enhanced phase change materials (NEPCMs) that work better [6–8]. These improvements enable more effective energy release and storage, which is essential for uses such as solar energy systems and building temperature control [9]. PCM-based nanoparticles have also shown promise in biological applications, especially in the fight against bacterial infections [10]. This is because they can be precisely distributed and are compatible with living things. All things considered, the combination of nanotechnology and PCMs offers a lot of potential for improving energy efficiency and solving environmental issues [6,7].

Nanotechnology's special qualities and uses have the potential to completely transform cancer diagnosis and treatment. By increasing the sensitivity and specificity of cancer detection, nanoparticles improve imaging methods like MRI, CT, and PET in diagnosis, enabling earlier identification and better patient outcomes [11,12]. These nanoparticles can target specific cancer biomarkers, allowing for the non-invasive detection of trace amounts of cancer-linked proteins [12,13]. By delivering therapeutic chemicals precisely to tumor cells, nanotechnology in treatments allows for customized drug delivery systems that reduce adverse effects while protecting healthy tissues [11,13]. Furthermore, the regulated release of medications from nanoparticles lowers the frequency of administration and improves therapeutic efficacy [12]. Other emerging uses include immunotherapies that use the immune system to fight cancer cells and the use of nano enzymes to increase the efficacy of radiation therapy [12,14]. Ultimately, the application of nanotechnology in oncology is paving the way for less invasive, more personalized, and more effective cancer treatment [15].

Phase change materials (PCMs) improved by nanotechnology have enormous promise for a number of applications, most notably heat control and energy storage. Nanoparticles like graphene and carbon nanotubes significantly increase the thermal conductivity of PCMs, addressing the intrinsic drawbacks of conventional PCMs, which frequently show low heat transfer rates [16,17]. Functionalized graphene nanoplatelets, for example, can increase thermal conductivity by up to **140.88 %** while maintaining thermal stability during prolonged cycling [18]. PCMs can also have two-dimensional nanosheets added to them, like MXene and hexagonal boron nitride, to make them better at handling heat and fix issues like leakage [19]. These developments enable the efficient use of PCMs in diverse applications like electronics, solar energy systems, and construction, thereby promoting sustainability and energy efficiency across various industries [20].

Researchers have demonstrated that the introduction of nanoparticles, such as metal oxides and graphene nanoplatelets, addresses the intrinsic low thermal conductivity of PCMs, restricting their effectiveness in thermal energy storage applications [21–23]. For example, research shows that using nanoparticles to promote dispersion with surfactants can boost heat conductivity by as much as **122.26 %** [24]. Additionally, different concentrations of nanoparticles can improve heat transfer performance and optimize melting characteristics; copper oxide stands out as especially useful because of its advantageous thermal properties [25]. Furthermore, the use of surfactants contributes to the stability of these nanocomposites across various heat cycles, guaranteeing their usefulness in industries like solar energy, biomedicine, and construction [23,24]. All things considered, nanotechnology is essential to improving the performance and effectiveness of PCMs for sustainable energy solutions.

Because of their improved thermal characteristics, nanostructured enhance phase change materials (NePCMs) perform noticeably better in energy storage and release than conventional phase change materials (PCMs) [26]. On the other hand, NePCMs with nanoparticles have better thermal conductivity than their traditional equivalents, up to **160 %** greater [27]. For example, compared to conventional materials, hybrid

nano-enhanced PCMs (HNePCMs) have shown up to **18.69 %** more energy storage capacity and **20 %** shorter melting times [28]. NePCMs are appropriate for a variety of applications, such as solar energy systems and thermal management in electronics, because the incorporation of nanoparticles also improves thermal stability and dependability [29–31]. All things considered, the developments in NePCMs show promise for the future of thermal energy storage technology.

2. Nanotechnology in phase change materials

Nanotechnology offers a transformative approach to enhancing the thermal performance of phase change materials (PCMs), which are widely used in thermal energy storage applications. Traditional PCMs suffer from low thermal conductivity and limited heat storage capacity; however, the integration of nanoparticles significantly mitigates these issues, enabling broader and more effective utilization in sectors such as construction, biomedicine, electronics, textiles, and automotive systems [32,33]. Nanoparticles enhance PCM efficiency by improving thermal conductivity, accelerating charging/discharging processes, and increasing latent heat storage. For example, the incorporation of carbon nanotubes, metal oxides, and metallic nanoparticles like Cu and Ag has shown to substantially boost thermal transfer and stability [34].

Fig. 1 demonstrates how nanotechnology significantly enhances the thermal storage capabilities of phase change materials (PCMs) by integrating them into various building components such as roofs, windows, wallboards, and flooring. This integration aims to improve the overall energy efficiency of buildings [35]. At the core of this enhancement is nanotechnology's ability to manipulate materials at the atomic and molecular level, allowing for the design of novel nanostructures with customized thermophysical properties. By incorporating nanomaterials into PCMs thus forming nano-enhanced PCMs (NEPCMs) the materials exhibit improved thermal conductivity, higher energy storage capacity, and better phase stability [34].

These advancements make NEPCMs particularly suitable for passive temperature regulation and support for renewable energy systems. The inclusion of different nanomaterials plays a crucial role in performance optimization: silica and alumina enhance thermal conductivity and mechanical strength; graphene contributes to rapid heat transfer; and nano clays improve dispersion stability and phase change reliability.

Nanoparticles, with their high surface-to-volume ratios and diverse morphologies, provide broad utility across applications including solar energy storage, environmental remediation, and biomedical systems [36,37]. Their encapsulation into PCMs (via micro/nanocapsules) enhances thermal reliability, prevents leakage, and supports high-cycle stability. Nano-encapsulation also facilitates better integration into textiles, electronics, and building materials.

2.1. Definition and concept

Nanotechnology, which is defined as the manipulation of matter at the atomic, molecular, and super-molecular scale, is opening up a wide range of industries, including electronics, medicine, energy, and aviation, and it is very helpful in the creation of new materials [38]. As this technology improves, thermochemical storage methods have gotten a lot of attention. However, phase-change materials not only offer a better state-of-the-art, but they also have a higher storage capacity than practical thermal storage methods across a wide range of possible storage temperatures. Because of their lower melting temperatures and increased thermal conductivities, these PCM materials make it easier for the heat storage units to charge and discharge [39]. Thermal energy storage (TES) materials, with an emphasis on phase change materials in particular, have gained more attention since the energy crisis of the early 1970s [40] because of their capacity to store large amounts of thermal energy [41]. Phase change materials (PCMs) enable the storage of large amounts of latent heat during phase transition by storing excess energy for use during peak demand periods. This has the potential to

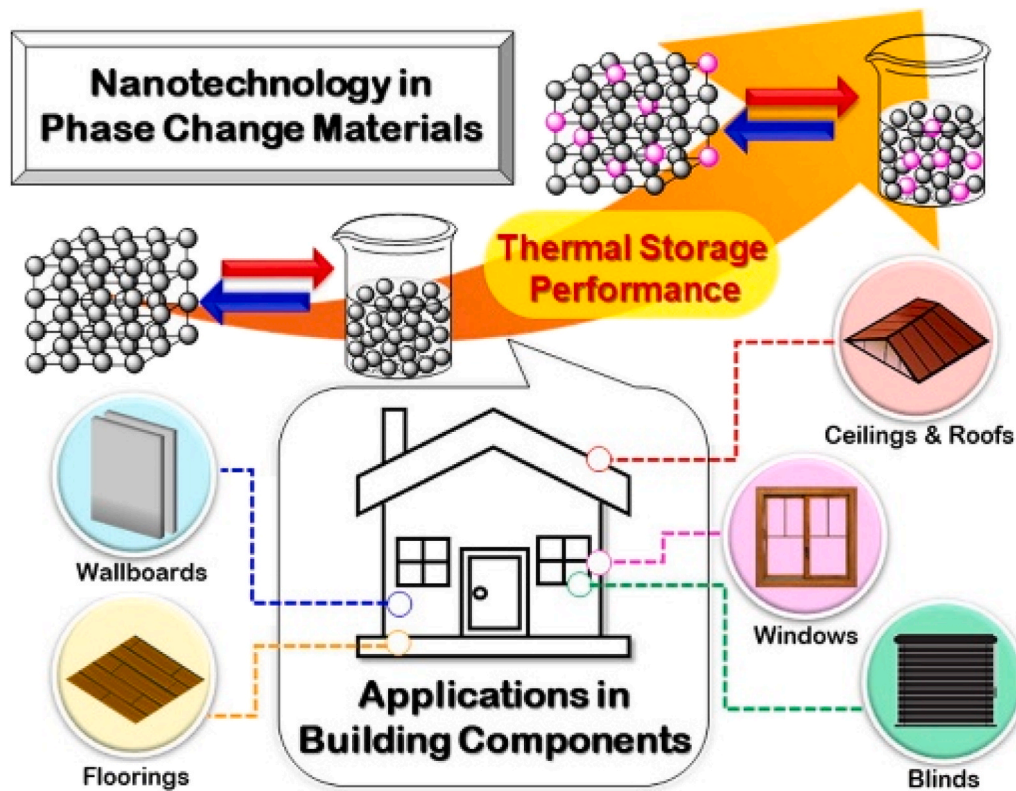


Fig. 1. Application in Building component [35].

both reduce overall energy demand through passive thermal regulation and increase the efficiency of renewable energies like solar power [42]. The intrinsic qualities of PCMs enhanced by nanotechnology are not a slight improvement but a significant one, increasing their stability, thermal conductivity, and thermal energy retention capacity. This greatly expands their suitability and adaptability for a wide range of applications [43]. Fig. 2 summarizes the desirable properties of effective PCMs: high latent heat, good thermal conductivity, high specific heat storage, mechanical/chemical stability, and minimal supercooling behavior [44].

2.2. Various nanomaterial types

Nanotechnology employs a variety of manufactured nanomaterials and nanoparticle types in at least one dimension [45–47]. Because

nanomaterials differ from bulk materials in a variety of ways, such as size, shape, surface area, and reactivity, they perform very well in a wide range of applications [48]. Carbon-based nanomaterials (NMs) are a specific family of nanomaterials composed of carbon atoms, including fullerene, graphene, nanodiamonds, and carbon nanotubes (CNTs) [49]. Due to their small size, comparable to many basic biomolecules, large specific surface area, high electrical and thermal conductivity, special optical qualities, and superior mechanical properties, carbon nanomaterials offer an endless array of customization and modification options, opening the door for a wide range of applications [50,51]. Using metallic nanomaterials with 25-nm Cu nanoparticles at 1 % weight on paraffin in a conductive environment can increase the rate of melting and solidification by about 33 % [52].

Nanostructure ceramic materials demonstrate thermal microscopy, differential scanning calorimetry (DSC), electrical, optical, magnetic, and refractory capabilities, as well as superior mechanical, corrosion, and chemical resistance. Researchers said that placing polymer nanoparticles in such micro-sized capsules improves protection in the stationary regime and reduces the heat conductivity of the coating [53]. Biomolecules, also known as biopolymers, differ from synthetic polymers in that they form polymeric structures through the covalent fusion of monomeric units. They have hardness at both normal and high temperatures [54], chemical inertness, and intricate, randomly configured architectural structures [55]. We examined the effects of nanoparticles on the thermal characteristics of microcapsules using scanning. Table 1 summarizes the various types of nanomaterials and their applications.

2.2.1. Nanoparticles

The most crucial general characteristics of nanoparticles, which change with size, include their surface area, optical qualities, homogeneity, fictionalization, quantum confinement, and other fractal-type features [69]. The first step in defining a nanoparticle is its size, which ranges from 1 to 1000 nm. We can develop effective methods for minimizing the formation of pollutants and lowering their emissions by

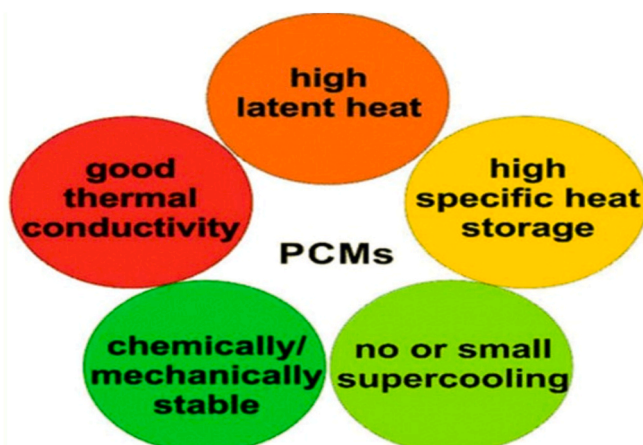


Fig. 2. Basic selection criteria of PCMs [44].

Table 1

Summary of the different types of nanomaterials and their applications.

Types of Nanomaterials	Used	Reference
Silica (SiO ₂)	Effects on the melting and freezing characteristics of phase change materials.	[56]
Metallic	Electrical and thermal conductivity.	[57]
Carbon Nanotubes	Enhance thermal conductivity and mechanical strength.	[58]
Nanotubes	Can improve the mechanical properties and thermal performance	[59]
Copper Oxide (CuO)	Improve heat transfer performance during phase change processes.	[60]
Alumina (Al ₂ O ₃)	Enhance thermal properties in composite phase-change materials.	[61]
Silver (Ag)	Added to the composite for its potential advantages in thermal conductivity.	[62]
Carbon-based	Enhanced Electrical Conductivity.	[63]
Metal Oxide	Used in coatings.	[64]
Graphene	Used in various nanocomposite applications.	[65]
Nano clays	Used in packaging materials and coatings.	[66]
Lithium bromide	Improving solar absorption refrigeration.	[67]
Nanofibers	Can be incorporated to increase surface area and improve heat transfer.	[68]

understanding the dynamic processes of nanoparticle formation and growth, such as in the combustion system. Nanotechnology holds enormous promise for improving the quality of the air, water, and soil in the environment [70–72]. In addition to having a wide range of morphological shapes, including spheres, cylinders, discs, platelets, hollow spheres, tubes, etc., nanoparticles also have a wide range of chemical forms, including metals, metal oxides, semiconductors, polymers, carbon materials, organics, and biologicals. One can create it using a variety of synthetic techniques based on gas, liquid, or solid phase approaches [73]. The applications of nanoparticles are numerous and include medical treatments, the production of solar and oxide fuel batteries for energy storage, and their widespread integration into a variety of everyday materials, like clothing, cosmetics, optical devices, catalytic, bactericidal, electronic, sensor technology, biological labeling, and the treatment of certain cancers. Recent years have seen a significant increase in interest in nanoparticles due to their remarkable qualities, which include outstanding resistance to oxidation, high thermal conductivity, and antibacterial activity [74].

Nanoparticle-enhanced phase change materials (NEPCM) are highly advantageous over traditional PCMs (i.e., PCM without nanoparticles) due to their higher rate of heat release and higher thermal conductivity. Reports indicate that NEPCMs present a significant potential for energy storage [75,76]. The dispersion of nanoparticles into PCM has made it feasible to accelerate unstable processes, enhance performance, and raise charging rates [77]. When 0.165 % Cu nanoparticles are present, the thermal charge duration decreases by 25.3 %, and when Cu-Al₂O₃ hybrid nanoparticles are present at a concentration of 0.165 %–0.816 %, the duration decreases by 10.8 % [78]. Copper nanoparticles guarantee that PCM starts storing and releasing heat at the proper time, leading to enhanced and improved heat transfer by reducing the melting duration by at least 10 % [78]. Researchers also discovered that adding 3 % of nanoparticles does improve the melting acceleration rate. At an inlet air temperature of 45 °C, the use of nanoparticles with PCM leads to a power savings of 7.18 % when compared to PCM alone. Using Cu nanoparticles with the PCM at a concentration ratio of 5 % results in a 7.41 % power savings by lowering the output air temperature and raising the coefficient of performance [79].

2.2.2. Composites at the nanoscale

Nanocomposite possess a high density of interfaces, which modify characteristic length scales and play a critical role in determining their physical properties [80,81]. Over the past two decades, significant research efforts have led to the development of nano-enhanced phase

change materials (NePCMs), a class of nanocomposite PCMs that incorporate carbon-based nano additives. Although the primary objective of these additions is to enhance thermal conductivity, substantial overall improvements in thermal performance have been achieved [82]. Graphene, graphene nanoparticles, and other two-dimensional carbon-based nanostructures exhibit nanoscale thickness and lateral dimensions, making them particularly attractive for thermal applications [82,83].

At the nanoscale, heat transfer in solid materials is governed by acoustic phonons and electrons [84]. In metallic systems, electron contributions to thermal conductivity (κ_e) dominate, accounting for approximately 98–99 %, while phonon contributions typically represent only 1–2 %, owing to the high density of free electrons [84,85].

Nanoscale confinement of PCMs allows for efficient encapsulation within extremely small volumes, resulting in improved stability and enhanced thermal performance [86]. PCMs can be effectively confined at the nanoscale through several approaches, including core-shell, longitudinal, interfacial, and porous confinement techniques. These strategies have been comprehensively reported in recent literature and offer promising routes for improving thermal behavior [86]. Nanotechnologies inherently enhance PCM performance by inducing pronounced changes in thermophysical and physicochemical properties at the nanoscale [87,88].

2.2.3. Capsules of nanoparticles

Researchers found that when the amount of polymerisable emulsifier was 2 % weight, nanocapsules achieved the highest level of encapsulation efficiency, chemical stability, and heat storage capacity [89]. Researchers synthesized phase change material (PCM) comprising n-octadecane (n-ODE) and n-eicosane (n-ESE) to create the nanocapsules (n-ODE@CLPS and n-ESE@CLPS) [90,91]. We placed a layer of nanocapsule PCM on the surface to enhance latent heat and achieve PCM encapsulation [92]. Phase-change micro-nanoencapsulation technology can resolve mismatched heat supply and demand [93]. Researchers believe the produced nanocapsules can store significant energy for use in fibers [94]. We used vibrant stirring to create microcapsules and nanocapsules containing 70 % n-octadecane [94]. The mass ratio of TEOS to MMA affected the structure of the nano capsules [95]. TEOS/MMA mass ratios of 1:3.5 and 1:5.75, respectively, produced the resulting spherical, homogeneous nanocapsules with a diameter of around 120 nm [95]. Researchers found that the nanocapsule particles were remarkably stable, even after 200 phase-change cycles, with no visible breaks and an enthalpy of melting of 181.5 J/g with very little loss [96]. Fig. 3 shows a nanoparticle size distribution curve, peaking around 90–100 nm. Over 22 % of particles fall within this narrow range, indicating good uniformity. Such nanoscale consistency is critical for optimizing PCM thermal performance [97].

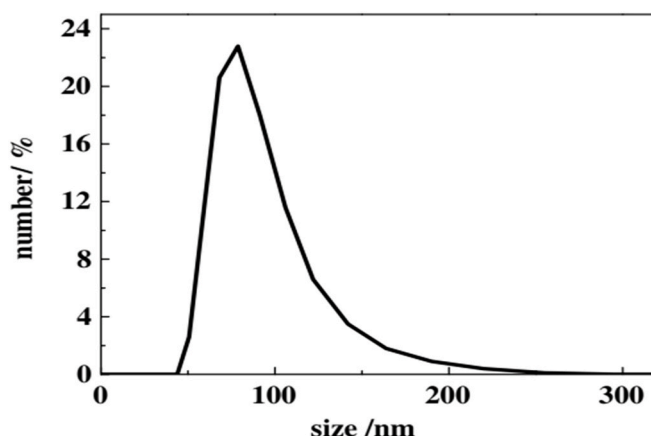


Fig. 3. Size distribution of the nano encapsulated PCM [97].

Additional advantages of nanoencapsulation include increased sustainability, stability, and loading capacity [98,99]. The readily broken nanoencapsulated phase change materials, or NanoPCM, have the ability to increase fluid viscosity during fluid movement [100]. Nanoencapsulated phase change materials (NanoPCM) are a popular choice for thermal energy storage materials because of their small size, large specific surface area, and quick heat transfer rate [100]. The nanocapsules possess excellent thermal stability and dependability [101]. Additionally, nanocapsules are believed to be a practical and effective means of simultaneously enhancing heat conductivity and preventing leakage [101]. Phase-change nanocapsules not only speed up the rate at which LFTF (Latent Functional Thermal Fluids) transfer heat, but they also make this material useful in more situations [102]. In a wide range of industries, such as electronics, medicines, textiles, cosmetics, chemicals, and construction, nanocapsules provide a number of solutions to issues [103]. Nano-capsules may hold a wide range of substances, including bacteria and energy-saving materials, due to their stiffness, anti-corrosive properties, and waterproof nature [35,44,104–106]. Table 2 presents influence of different nanomaterials on thermophysical properties of PCMs.

3. Developments in PCM nanotechnology

A real-world building application enhances the performance and compatibility of PCMs [123]. Applying nanotechnology to various building components in conventional PCMs can yield high-performance nanocomposite materials [123]. The development of thermal fluids enhanced by nanotechnology for energy system optimization boosts conversion process efficiency and has broad economic, environmental, and social impacts [124]. Nanotechnology has greatly aided in enhancing the radiative cooling capabilities of smart fabrics [125].

The interactions between nanoparticles and phase change materials (PCMs) at the molecular level are complex and significantly influence the thermal properties of the resulting nanocomposite materials. Molecular dynamics simulations have been instrumental in elucidating these interactions. For instance, the integration of nanoparticles such as SiO₂ into molten carbonate PCMs results in an initial increase in specific heat capacity (SHC), attributed to the separation of amorphous carbonate molten cations into different components, which alters the distribution of electrostatic and long-range interactions, thereby enhancing SHC [126]. Similarly, the addition of nanoparticles like Al₂O₃, CuO, and SiO₂ to chloride salt PCMs improves thermal performance by increasing the average potential energy of the atoms, which strengthens interatomic interactions and enhances heat transfer and storage capabilities [127]. However, these enhancements come with trade-offs, such as increased viscosity, which can affect the material's flow properties [127].

The molecular-level interactions are further complicated by the potential for nanoparticles to cause localized conformational changes in proteins or other organic components within the PCM matrix, as seen in studies of protein-nanoparticle interactions where proteins may undergo partial unfolding to maximize contact with the nanoparticle surface [128]. Additionally, the thermodynamic interactions between

nanoparticles and cellular membranes highlight the potential for nanoparticles to disrupt cellular processes, which underscores the importance of understanding these interactions for both material performance and safety considerations [129]. The concentration of nanoparticles also plays a crucial role, as varying concentrations can significantly alter the mass fraction and melting characteristics of nano-enhanced PCMs, with materials like CuO showing particular efficacy in enhancing melting due to their favorable thermal properties [122]. These insights collectively guide the design and optimization of nanoparticle-PCM systems for improved thermal energy storage applications.

3.1. Improved heat conductivity

In addition to increasing energy efficiency, phase change materials (PCMs) can greatly improve interior thermal comfort in building applications [130]. The introduction of highly conductive metallic and carbon-based nanoparticles, along with expanded graphite, metallic foams, and PCM encapsulation, all contribute to the improvement of thermal conductivity [131]. One of PCM's main challenges in the field of latent heat storage applications is improving thermal conductivity [132]. Nanoparticles can significantly increase the thermal conductivity of the base PCM [133].

3.1.1. CNTs, or carbon nanotubes

Carbon nanotubes, possessing the strongest strength of any material, the highest thermal conductivity, and remarkable field emission qualities, are considered the ultimate form of carbon fibres [113]. Recently, carbon nanotubes (CNTs) have gained recognition for their exceptional qualities, which outperform those of any other material [114]. At distances between 3.8 Å and 8.0 Å, the PCMs are seen to preferentially organize themselves in one or two ring-shaped layers that are clearly distinct from the CNT walls [115]. Carbon nanotubes, or CNTs, have garnered significant interest due to their hollow structure, making them ideal for filling or encasing PCMs, and large specific surface area. They also have outstanding physical qualities, including high electrical conductivity, low density, and excellent strength [115]. We stirred CNTs in concentrations of 1, 2, 5, and 10 wt percent into liquid wax at 60 °C to create the composite phase change materials (PCMs). According to the experimental results, the composite PCMs' thermal conductivity increases as the amount of CNTs increases [116]. Another lightweight nanomaterial that shows promise for heat transfer applications is carbon nanotubes (CNTs) [116]. Researchers have studied CNTs as fillers to increase the thermal conductivity of PCMs [116]. The temperature differentials fell by 92.0 % and 12.2 %, respectively, along the length and depth of the composite shelf that contained 3 %-CNT composite PCMs [117]. There are two types of carbon nanotubes: single-walled (SWNTs) and multi-walled (MWNTs). MWNTs have a thermal conductivity of about 3000 W m⁻¹ K⁻¹, and SWNTs have a thermal conductivity of about 6600 W m⁻¹ K⁻¹ [118]. Since carbon nanotubes (CNTs) thermal conductivity ranges are from 2000 to 6000 W/m K, they are very useful for making PCMs better at storing and releasing heat [133]. The incorporation of amphiphathic CNTs with high thermal conductivity positions

Table 2
Influence of different nanomaterials on thermophysical properties of PCMs.

Nanomaterial type	Typical loading range	Thermal conductivity enhancement	Latent heat variation	Scientific inference	References
Graphene/graphene nanoplatelets	0.1–5 wt. %	~80–160 % increase	–5 % to –15 %	High aspect ratio enables percolation networks; dispersion quality dominates performance	[82,83, 107–112]
Carbon nanotubes (CNTs)	0.5–10 wt. %	~60–150 % increase	–5 % to –20 %	Conductive pathways reduce melting time; agglomeration at high loadings	[113–121]
Metal oxides (Al ₂ O ₃ , CuO, TiO ₂ , SiO ₂)	0.1–5 wt. %	~20–120 % increase	–5 % to –15 %	Micro-convection and heterogeneous nucleation enhance heat transfer	[21–25,122]
Hybrid nanoparticles	0.1–3 wt. %	~100–160 % increase	–5 % to –12 %	Synergistic effects improve heat transfer efficiency	[27–29]
Nano-clays/silica	1–10 wt. %	~20–60 % increase	–5 % to –10 %	Improved shape stability and leakage resistance	[56,64–66]

these PCMs as attractive candidates for a variety of applications, including self-cleaning materials, electronic device thermal management, and energy-efficient buildings [119]. In addition to increasing PCMs' thermal conductivity, CNTs also aid lower cooling rates at particular temperatures and increase the PCM's stability range [120]. To increase the thermal conductivity of phase transition materials, researchers have also looked into employing carbon-based materials like graphene, carbon nanotubes, and carbon nanofibers [121].

3.1.2. Graphene oxide and graphene

The basic component of 3D graphite, 1D carbon nanotubes, and 0D fullerenes is graphene. Scientists have paid close attention to it because of its unusual planar structure and special electrical characteristics [134]. Sp² bonds between carbon atoms create graphene [135]. Conceptually, a single layer of six-membered rings with a conjugated structure makes up graphene, a planar aromatic macromolecule [136, 137]. Graphene and its composites have become cutting-edge biomaterials in the field of nanomedicine, with tremendous promise for a variety of uses, including drug delivery nanocarriers, next-generation biosensors, and biological and cellular imaging probes [136]. We functionalized graphene oxide (GO) nanoparticles with ethoxytrimethylsilane (ETMS) to enhance their stability and dispersion in paraffin-based PCMs [107]. The addition of graphene and graphene oxide to paraffin significantly increases the thermal conductivity of the resultant PCM [107]. Researchers have extensively produced phase transition material using graphene, graphene oxide, and its derivatives [107]. It is known that graphene and graphene oxide are useful nanomaterials for making PCMs better at storing energy and withstanding heat. However, if the matrix is not spread out correctly, it can cause structural flaws in the composite and make it work less well than expected [107]. Nanomaterials based on graphene are renowned for their remarkable mechanical strength and ease of functionalization in a wide range of applications [108]. By nature, almost all gas molecules cannot flow through graphene nanosheets [109]. We employ graphene nanoplatelets (GNP) and graphene oxide (GO) to enhance thermal conductivity and decrease supercooling, respectively [110]. Graphene oxide (GO) is hydrophilic because it has a lot of functional groups with oxygen on its surface [111]. This lets it make strong connections with polymers through covalent and hydrogen bonds. We conduct a review on the application of small graphene-based structures in thermal management, energy storage, and electromagnetic interference (EMI) shielding [112].

3.2. Better kinetics of phase change

Kinetic studies and thermal stability are very important for switching in high-temperature areas of nanoscale phase-change materials that will be used in the next generation of non-volatile storage class memory [138]. The kinetic and thermodynamic characterization of the degrading mechanism in PCM disposal and its nanocomposite of PCM helps us understand the complex chemical reactions at play and gives us information on the thermal parameters for large-scale operations [139,140]. The melting kinetics of both bulk and nanoconfined C20 demonstrate a decrease in the effective activation energy inside the pores. However, due to the short timescales and small length scales involved, the crystallization kinetics at high temperatures, which serve as the rate-limiting component during switching, encounter significant challenges [141].

3.2.1. Catalysts at the nanoscale

Industrial nanoscale catalysis makes extensive use of materials containing a range of nanostructures [142]. Researchers first proposed this concept a century ago, seeking to control the chemical identity, local environment, and electrical properties of active sites on catalytic surfaces to optimize their reactivity in specific applications [142]. Production of Composites Macroscopic, three-dimensional structural materials, possessing versatile properties, serve as excellent substitutes [143]. 3D structural materials, including carbon foams, metallic foams,

graphene aerogels, and multipurpose porous scaffolds [143]. The relationships between properties (heat conductivity and form stability) and topologies in 3D structural composite PCMs [143,144]. A few nanometers are the size of very small transistors, graphene and carbon nanotubes, memory components, light sources, motors, sensors, lasers, and pumps [145]. For carbon and carbon bonds, the graph's minimum length is around 0.142 nm [145]. Enzymatic catalysis is the most efficient and ecologically benign type found in nature [146]. The two main determinants of nanoparticle catalytic efficacy are (i) meticulously controlled NP production and (ii) a molecular approach to find more specific NP applications [146]. A suspension or solution of nanoparticles in a fluid, including a nano catalyst, is a popular definition of a homogeneous system [146]. Advances in the creation of nanostructures and devices using biomolecules, colloids, surfactants, and self-organizing polymers are covered in the study [147].

Since the amount of energy produced by the sun and wind varies according to environmental changes, thermal energy storage (TES) technologies can help provide reliable and sustainable energy [148]. Heat transfer, phase transition, and capillarity performance in micro-/nano confinement should differ significantly from the bulk phase, where significant theoretical and experimental research has been carried out for more than a century [148]. A few problems prevent the use of inorganic multifunctional nanomaterials for enhanced energy applications. These issues, which include investigating creative structural material systems, identifying hitherto unidentified features of existing structures, and enhancing the performance of current materials, are mostly brought about by the study of novel materials [149].

A last section is offered for a variety of uses. The most researched applications soft electronics, sensors, energy, catalysis, nanomedicine, and biological applications are covered in the main subsections [150]. A developing class of electronics known as soft electronics which can operate under mechanical deformation is required for applications such as soft robotics, wearable electronics, artificial skins, soft junctions, and more [150]. The high cost of the nanoparticles and the uncertain cost-benefit analysis are the main obstacles to the implementation of PCM-NEPCM systems. NEPCM is a preferred option for thermal fluids, energy storage, heating/cooling, and waste heat recovery [151]. Nanofluids, a type of liquid that contains nanoparticles, have been found to have superior thermal, optical, and electrical properties when compared to traditional fluids. However, PCMs are ideal for a range of thermal applications because they are materials that can store and release thermal energy through a phase transition process [151].

3.2.2. Encapsulation techniques

Depending on the type of supporting material and the chemical properties of the OPCMs, a variety of methods are available for their encapsulation [152]. Every encapsulating technique has advantages and disadvantages [152]. Due to the expansion of the global economy and the consequent rise in energy consumption, there has been a recent surge in the search for more energy-efficient technology. The environment has suffered as a result of this expansion [153].

Given the present global concern over CO₂ emissions, PCMs have the potential to store a substantial quantity of energy, whether incorporated into building materials or used directly as passive systems [154]. The food and textile sectors, smart electronics, home concentrated solar hot water systems, and the construction sector have widely used iPCMs in recent years [154,155]. Around the world, people widely use conventional resources due to their affordability and ability to meet energy demands. However, these resources also come with disadvantages, including habitat destruction, ozone layer depletion, air and water pollution, and climate change. Research indicates that PCMs exist in various forms, such as liquid metals and inorganic salts, and find application in diverse contexts such as spacecraft architecture and residential hot water tanks [156].

Despite numerous studies examining their thermophysical properties, few researchers actually exploited NEPCMs for specific purposes

[157]. For instance, researchers simulated a thermal cell using fins with cylindrical shapes, which functioned as a heat sink and a thermal conduction booster for a thermal aluminum sheet [157]. Differential scanning calorimetric studies with changing rates of heating and cooling can give useful information about PCM, like the exact temperature of the phase transition and the latent heat of fusion [158]. There are several applications, including air conditioning, cold packaging, food storage, medicine, and free building cooling [158]. This integration's main objectives are to increase any thermal system's efficiency, reduce a building's cooling load, and employ passive cooling in conjunction with heat recovery [159–162].

The comparison of encapsulation techniques for phase change materials (PCMs), such as in-situ polymerization and layer-by-layer assembly, reveals significant differences in efficiency, cost, and impact on PCM performance. In-situ polymerization is noted for its ability to create robust microcapsules that enhance thermal stability and reduce leakage, making it a cost-effective choice for high-temperature applications [163, 164]. Conversely, layer-by-layer assembly offers precise control over the thickness and composition of the encapsulating layers, which can improve the thermal conductivity and overall performance of PCMs, albeit often at a higher cost due to the complexity of the process [165]. Microencapsulation techniques, including both methods, have been shown to significantly enhance the thermophysical properties of PCMs, such as heat transfer efficiency and mechanical strength, thereby expanding their applicability in thermal energy storage systems [166]. Ultimately, the choice of encapsulation technique should be guided by specific application requirements, including temperature range and desired performance characteristics [167].

3.3. Increased stability and longevity

The long-term stability of the material, or its service life, determines the successful application of Latent Heat Storage (LHS) materials. The material's properties and its ability to store latent heat must stay stable during this time, and thermal cycling or interactions with phase-change materials and storage systems [157] must not change them. The main barriers to the widespread adoption of LHS currently include the durability of heat exchanger tubing and the longevity of PCM containers [168]. The storage phase change material (PCM) and the heat transfer fluid (HTF) exhibited inadequate heat exchange, thereby prolonging the

charging and discharging durations of the PCM. Yuksel et al. proposed a model [168]. Many researchers have developed nanophase change materials (PCMs), characterized by the assessment of their thermophysical properties. A summary of key methods for characterizing nanoPCM is available [169]. Ensuring the stability of nanoparticles within phase-change materials is essential for the long-term functionality of nanocomposite materials. [169]. The review is divided into five main parts: an introduction that talks about the history of emulsions and how to make microcapsules; a list of PCM microcapsules and emulsions, both those that are being studied and those that are already on the market; a study of the thermophysical and rheological properties of PCM emulsions and mPCM slurries, including subcooling and fluid stability; a look at heat transfer phenomena in the fourth section; and finally, a list and description of some uses for these new fluids [170]. Optimizing multiple parameters is essential in the pre-design phase, and predictions from numerical simulations can be valuable for assessing overall performance [171]. This section presents the current advancements in lithium-ion battery thermal management and identifies potential avenues for future research and enhancement [172]. Fig. 4 illustrates various graphene-based phase change composites (PCCs). These include aerogels, microcapsules, and functionalized graphene-enhanced PCMs. Graphene improves thermal conductivity and structural stability of PCMs [161].

Degradation processes such as oxidation and phase separation significantly impact the long-term stability of nanocomposite phase change materials (NePCMs). Oxidation is a primary factor leading to the deterioration of materials, as it alters their structural and chemical properties, ultimately affecting their performance and longevity [121]. The presence of fillers and the composition of polymer blends can influence degradation rates, with certain additives like antioxidants showing promise in mitigating oxidative effects [110]. Additionally, phase separation, which occurs when a system loses stability, can further compromise the integrity of NePCMs, leading to reduced thermal storage capabilities [112]. The stability of these materials is crucial, especially under repeated thermal cycling, as degradation can result in significant losses in latent heat storage potential [139]. Therefore, understanding and addressing these degradation mechanisms is essential for enhancing the durability and reliability of NePCMs in thermal energy applications.

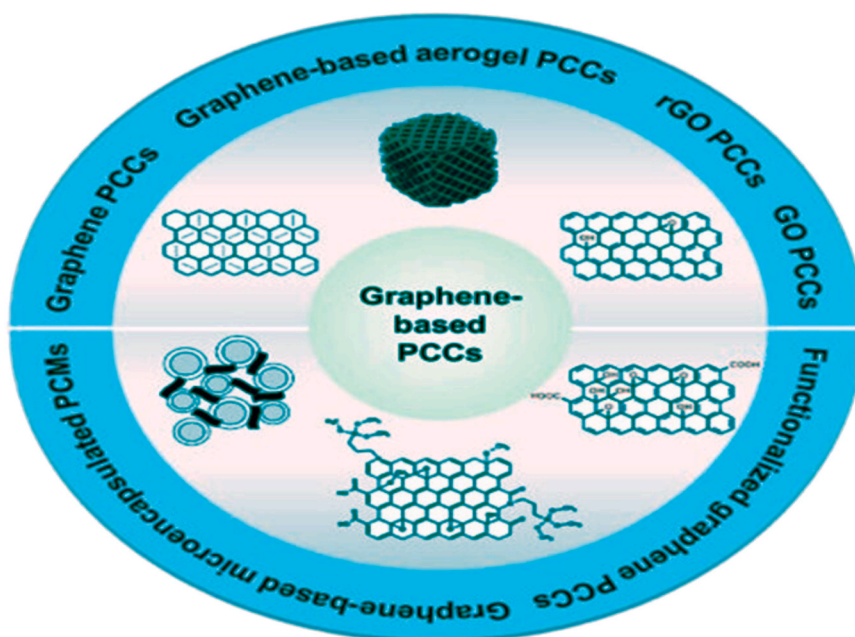


Fig. 4. Overview of graphene-based PCCs covered in this work [161].

3.3.1. Silica nanocapsules

We can categorize phase change materials (PCMs) into several classes, such as solid-liquid, liquid-gas, and solid-solid types [173]. A process that enhances thermal conductivity, mitigates leakage during phase transitions, and potentially facilitates interaction with the surrounding matrix can effectively encapsulate Phase change materials (PCMs) [173]. This document presents a summary and analysis of the three main preparation techniques: sol-gel polymerization, in-situ polymerization, and easy self-oxidation [174]. Phase change materials (PCM) possess a high specific heat during the process and are characterized by their capacity to store substantial thermal energy during phase transitions, typically between solid and liquid states, at a specific temperature [175]. This represents the first comprehensive analysis of inorganic PCM encapsulation methods [175].

We highlight the necessity of developing structurally advanced and multifunctional composite phase change materials (PCMs), which have potential applications in air conditioning, photothermal and electromagnetic-thermal energy storage and conversion, smart textiles, portable electronics, battery thermal management, and intelligent drug control and release [176]. Improving energy storage technologies for renewable sources represents a significant and promising approach to tackle the current energy crisis and global environmental issues [177]. Supercapacitors store charges exclusively on their surfaces, which allows for high power delivery due to the absence of diffusion processes that could hinder charge storage [177]. Nanocrystals with a high surface energy have an open surface structure and a lot of step and kink atoms that aren't coordinated well [178].

Thermal energy systems can store latent heat for use in industrial and residential applications during off-peak periods. Phase Change Materials (PCMs) exhibit a high enthalpy for heat storage, resulting in significant thermal energy storage capacity. Applications for them include under-floor heating systems, solar energy, textiles, refrigeration, electronic cooling, waste heat recovery, and food and milk preservation [179]. NePCMs exhibited effective system performance in solar thermal energy storage systems (TESS). The volume concentration of the nanoparticles is significant [179]. Currently, various methods exist to refine these properties to varying levels of atomic precision. Progress relies on the availability of analytical methods that can elucidate relevant structural characteristics. This task is challenging due to the intrinsic complexity and dynamic effects often present in even the most basic heterogeneous catalysts under reaction conditions [180]. Using the latent heat that is created when a phase change material (PCM) changes from solid to liquid, latent heat storage (LHS) is an effective way to control temperature. This subject has attracted interest owing to its several advantageous characteristics, such as high repeatability, stable heat transfer at a constant temperature, and the capacity to release and absorb substantial heat within a limited temperature range [181]. Recent advancements in nanotechnology have attracted global scientific interest due to its potential applications across diverse fields, as well as the building and food industries [182].

3.3.2. Polymer-based nanocomposites

Analyzing the intercalation process between nanoparticles and polymer bases is essential to achieve the desired mechanical, thermal, optical, and electric characteristics [183]. The primary challenge of PNCs lies in the complex regions where nanoparticles interact with polymer matrices [183]. PNCs have garnered significant attention from researchers addressing ecological and energy-related challenges. PNCs have more benefits because they contain nanoparticles of similar lengths and polymeric supports, as well as a large interfacial area compared to the volume of the material [184]. The nanocomposites use the conducting polymer's strong pseudocapacitive properties through a double-layer capacitor and the nanoparticles' unique mechanical properties [184]. The high storage energy density per unit volume/mass and the reduced temperature difference between heat storage and release make PCM-based latent heat storage (LHS) preferred over sensible

energy storage [185].

Nearly all technological fields are presently directing their research and development initiatives toward nanotechnology [185]. Initial findings regarding exfoliated clay, along with recent studies on carbon nanotubes, carbon nanofibers, exfoliated graphite (graphene), nanocrystalline metals, and various nanoscale inorganic fillers or fiber modifications, have generated significant interest in polymer matrix-based nanocomposites [185]. The operating temperature of a photovoltaic panel and the solar energy it receives influence its efficiency. The solar panel's temperature rises. The increase in environmental pollution, climate change, ozone depletion, and policies that endorse the continued use of fossil fuels represent significant challenges facing the world in the twenty-first century [186]. Thermoplastic heat exchangers offer advantages over traditional metal heat exchangers, including lower cost, enhanced durability, and improved resistance to corrosion and chemical degradation [187]. Consequently, they can reduce total energy consumption while enhancing the efficiency of building HVAC systems [188].

Polymer nanocomposites exhibit a diverse range of properties that no other class of conventional materials can match [189]. Nearly all technological fields are currently emphasizing research and development in nanotechnology. We now recognize that crystallization in restricted regions influences both the rate and extent of crystallinity [186]. The family of materials referred to as polymer nanocomposites (PNCs) exhibits remarkable properties [183]. Additionally, a direct correlation exists between the advancement of nanotechnology in the form of polymer nanocomposites (PNCs) and environmental sustainability. The case of biopolymer nanocomposites and their applications examine this relationship [183,190,191]. Fig. 5 shows a 2:1 layered clay structure composed of tetrahedral-octahedral-tetrahedral sheets. Sodium ions (Na^+) are located in the interlayer gallery, allowing ion exchange and intercalation. Such structures are key for nano clay-based PCM composites, enhancing thermal and mechanical properties [186].

Table 3 illustrates the utilization of advancements in nanotechnology for phase change materials (PCMs).

4. Applications of nanotechnology-enhanced PCMs

4.1. Building and construction

Global energy consumption is increasing significantly as a result of population growth, with approximately 200,000 individuals added each day. Renewable energy has the potential to decrease consumption by 20–25 %. Furthermore, advanced materials used in construction and finishing, especially in residential buildings, present further opportunities for reducing electricity usage [199]. Studies show that in the European Union and other developed countries, buildings represent approximately 40 % of total energy consumption and contribute nearly 30 % of annual greenhouse gas emissions [200]. Heating, ventilation, and air conditioning (HVAC) systems account for approximately 50 % of this consumption. Enhancing energy efficiency in the built environment is crucial for reducing global energy consumption, addressing climate change, promoting environmental sustainability, and strengthening energy security [201,202]. TES systems incorporating PCMs improve thermal mass and comfort while decreasing energy consumption and facilitating renewable energy use by mitigating temperature fluctuations within buildings [203]. Fig. 6 illustrates the use of solid-liquid and solid-solid Phase Change Materials (PCMs) in buildings to regulate temperature through walls, roofs, floors, and other elements [204].

Conventional phase change materials (PCMs) have incorporated nanotechnology to prevent leakage, enhance thermal conductivity, accelerate the phase change process, and optimize latent heat utilization. Additionally, doping PCMs with nanoparticles such as copper and carbon nanostructures further enhance thermal performance and produces high-performance nanocomposites for practical building applications across various components [205,206]. Fig. 7 demonstrates

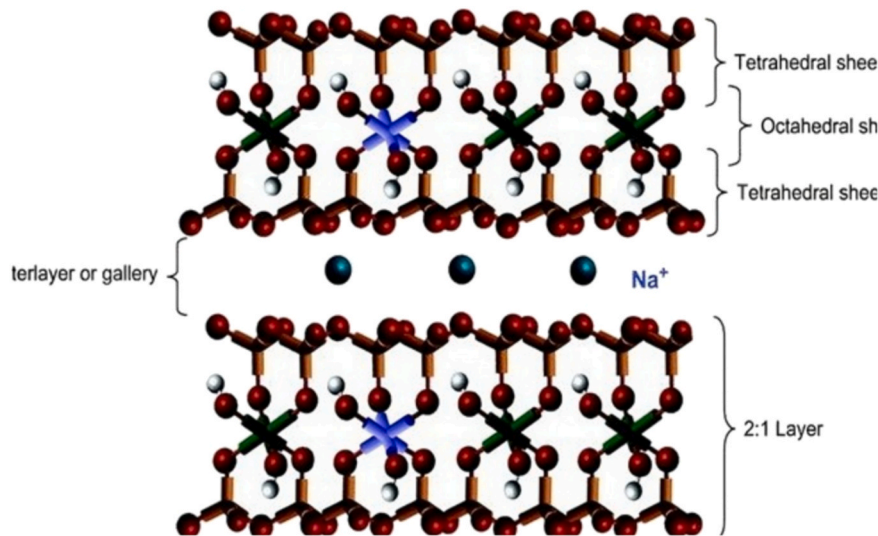


Fig. 5. Structure of sodium montmorillonite [186].

Table 3

The application of advances in nanotechnology for PCMs.

Key Aspects	Applications
Enhanced Nucleation	During the phase transition process, crystal formation can be aided by the nucleation sites that nanoscale catalysts can provide. This may result in more rapid and regulated rates of melting and solidification [192].
Latent Heat Storage	Building heating and cooling systems, solar thermal power plants, and electric car batteries may all benefit from the efficient storage and release of thermal energy provided by PCMs with integrated nanoscale catalysts [193].
Temperature Control	In a variety of applications, including food preservation, medical devices, and temperature-sensitive manufacturing processes, these materials can be employed to maintain a consistent temperature [194].
Silica encapsulation	To increase their long-term stability, avoid phase segregation, and improve their thermal conductivity, PCMs are encased in silica shells. This method works especially well with high-temperature PCMs [195].
Carbon nanotube encapsulation	To increase their phase transition rate and thermal conductivity, PCMs are enclosed in carbon nanotubes. Applications like electronics cooling that call for quick heat transfer employ this method [196].
Medical applications	Medical devices that employ encapsulated PCMs for medication administration and temperature control [197].
Enhanced Durability	NEPCMs are more resistant to phase segregation, leakage, and degradation, ensuring long-term performance [198].

energy savings in buildings using PCMs, highlighting **15.7 %** savings from latent storage and **18.5 %** from sensible storage, improving overall energy efficiency [207].

4.1.1. Energy-efficient building materials

By integrating materials like mortar with thermal storage capabilities to modulate indoor temperatures, we can enhance building envelopes characterized by high energy consumption [208]. This is an effective way to redistribute energy. Latent heat and effective temperature regulation enable Phase Change Materials (PCMs) to possess significant thermal energy storage capacity. These properties render them appropriate for lightweight construction and non-load-bearing applications, contributing to reduced energy demand in buildings when incorporated into construction materials. While the integration of phase change materials (PCMs) into building materials is advantageous, it also presents challenges such as corrosion, leakage, and compatibility with

concrete [209,210]. Nanoencapsulation techniques [203] can address leakage issues. Regardless of their weight classification, building structures can incorporate nano-enhanced phase change materials to enhance both overall and localized thermal mass properties. HVAC systems can also employ them as centralized or distributed energy storage solutions, which aid in mitigating energy supply and demand imbalances through substantial peak load shifting and reduction, thereby enhancing performance [202]. Adding longer surfaces, fins, heat pipes, and thermally conductive nanoparticles to phase change materials (PCMs) is one way to make them better at conducting heat and storing energy. Buildings can utilize these nanoparticles to enhance the properties of phase change materials (PCMs) for both passive and active energy management systems [200]. Fig. 8 illustrates how Phase Change Materials (PCMs) are incorporated into building materials like cement, insulation, gypsum, and paints. PCMs help regulate indoor temperature, reduce energy consumption, and enhance thermal efficiency. Conditioned PCMs enable absorption, storage, and release of heat, leading to more efficient and adaptable building systems [126].

4.1.2. Smart windows and facades

Glazing units, such as traditional windows, glazed façades, and glass roofs, serve essential functions in buildings by facilitating natural light, ventilation, visibility, and passive solar heating. Nevertheless, these units generally exhibit lower thermal resistance in comparison to alternative building materials [204]. Windows represent the most significant thermal vulnerability between a room's interior and exterior. In cold climates, windows contribute to 10–25 % of heat loss from the interior to the exterior. In hot climates, increased solar radiation through windows elevates the cooling load on air conditioning systems [211]. The incorporation of phase change materials (PCMs) into glazing systems can significantly decrease energy consumption for heating and cooling, mitigate indoor temperature variations, and improve overall building energy efficiency. Even so, phase change materials (PCMs) can't be used in many situations because they don't conduct heat well and store and release energy slowly [212]. The incorporation of paraffin wax into double-glazed windows represents a novel strategy for minimizing energy usage and controlling indoor thermal conditions. The low thermal conductivity of paraffin wax restricts its efficacy. Improving thermal performance through the incorporation of nanoparticles into phase change materials (PCMs) presents a viable solution [213]. The integration of a hybrid nanoparticle-enhanced phase change material (Al₂O₃-CuO hybrid Nano-PCM) in windows enhances thermal inertia and insulation; however, its elevated photothermal conversion may lead

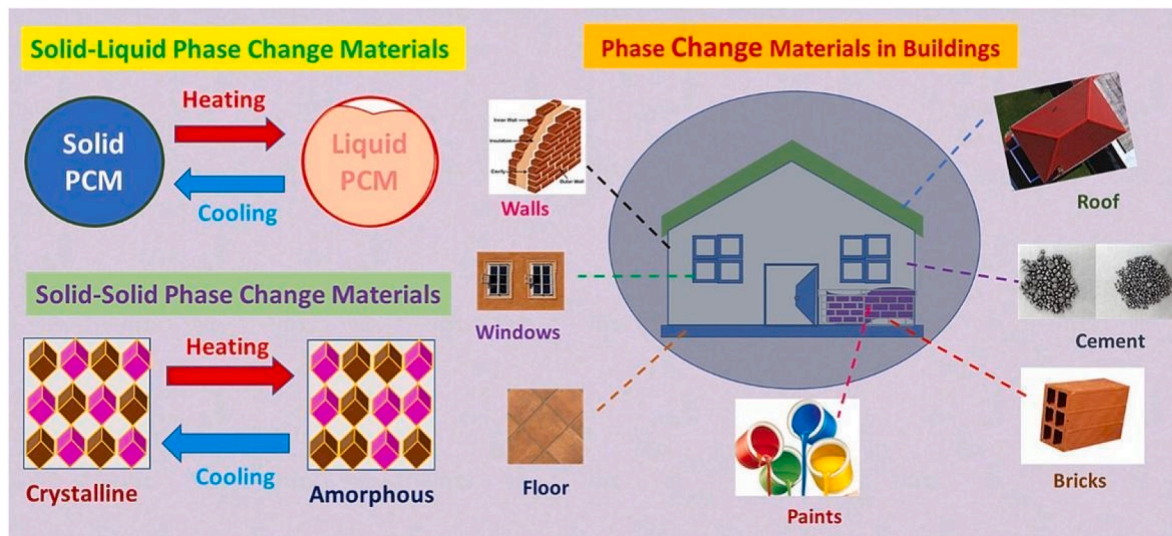


Fig. 6. Application of PCM in buildings[204].

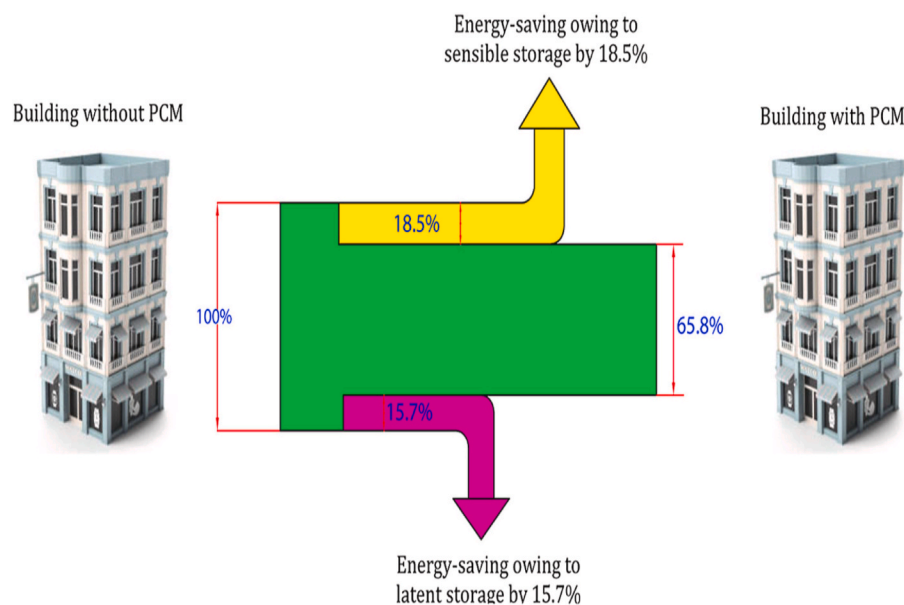


Fig. 7. Statistical review of world energy, 2021 [207].

to localized overheating during the summer, thereby diminishing thermal comfort. Adding hybrid nanoparticles lowers the amount of visible light that passes through liquid phase change materials (PCM). This could make it harder for natural light to pass through because PCM doesn't conduct heat well. Window units with nanoparticle-dispersed paraffin wax work better in terms of optics and heat transfer than those filled with pure paraffin. [214,215]. Adding nanoparticles to phase change materials (PCM) speeds up the melting and solidification processes, keeps the PCM in its liquid state longer, and raises the temperature of the glass's inner surface. When TiO_2 and ZnO nanoparticles are added to phase change materials (PCM), the temperature of the inside surface drops by 7°C and the total melting time drops by 1.5 min. This happens in the summer. [216]. Fig. 9 illustrates a PCM-assisted solar thermal wall system for energy storage. Incident solar radiation is captured, and a thermocouple monitors temperature changes to control PCM circulation. The pump and control unit regulate heat storage and release, improving thermal regulation in buildings [217].

4.2. Renewable energy storage

Renewable energy sources, including solar, tidal, wind, hydropower, and biogas, can sustainably fulfill global energy requirements, with solar energy demonstrating the highest potential [218]. Energy generation from renewable sources is contingent upon weather conditions and the reduction of costs associated with these sources, necessitating the implementation of energy storage solutions. Thermal energy storage in water tanks frequently results in energy loss and may require impractically large tank sizes for buildings. Thermal energy storage (TES) enhances energy efficiency by capturing thermal energy from intermittent heat sources [219–221]. Latent heat thermal energy storage, which utilizes phase change materials with high thermal conductivity, is an effective method for optimizing heat transfer and energy storage in solar energy applications [222]. Thermal Energy Storage (TES) extensively utilizes Phase Change Materials (PCMs) due to their significant thermal storage capacity and consistent phase transition temperature, making them suitable for renewable energy applications like solar energy

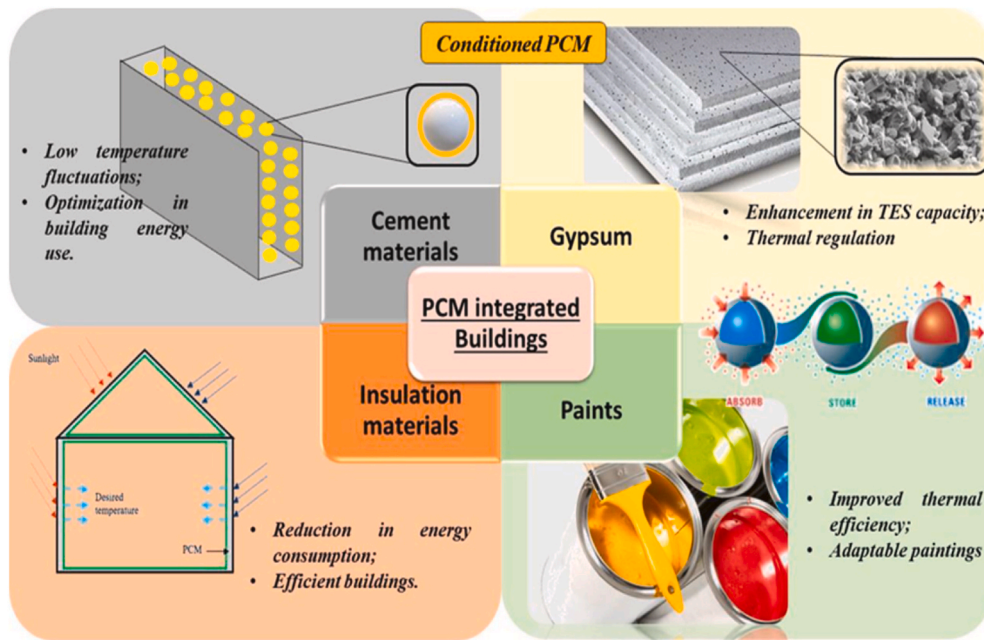


Fig. 8. PCM integrated buildings [126].

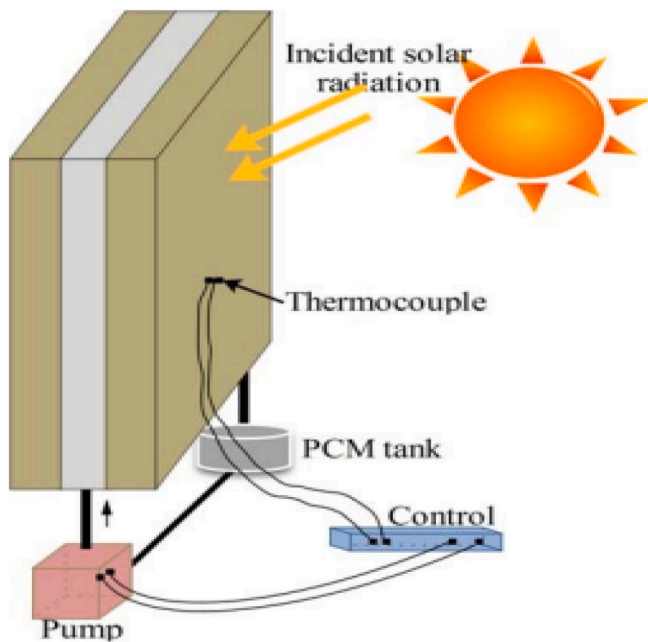


Fig. 9. PCM filled glass windows [117].

harvesting [223]. Phase Change Materials (PCMs) effectively address the disparity between energy supply and demand by storing and releasing energy in response to temperature variations. However, issues such as low thermal conductivity, instability, corrosiveness, supercooling, and leakage constrain their extensive application. Additives can improve thermal conductivity and minimize supercooling, whereas nano-encapsulation effectively mitigates PCM leakage [224].

4.2.1. Solar thermal energy storage

Recent years have witnessed substantial research advancements in solar energy conversion technologies, such as solar collectors, photovoltaic thermal systems (PV/T), solar water heaters, and desalination [199]. Solar energy represents a multifaceted renewable resource with

various contemporary applications. Efficient utilization necessitates a storage medium to capture surplus energy and release it as required [225]. Solar photovoltaic (PV) power generation and concentrated solar thermal power (CSP) represent the primary technologies for solar energy utilization; however, the industry is pursuing more cost-effective and efficient alternatives. Phase change materials (PCMs) represent viable alternatives, characterized by reduced costs and enhanced energy storage capabilities [226–231]. Fig. 10 highlights the role of PCM thermal energy storage in multiple domains. It connects climate change adaptation, renewable energy, and power outage mitigation with PCM systems. Stored thermal energy can support appliances, vehicles, and buildings for improved energy sustainability [127].

4.3. Textiles and apparel

Fabric, as the primary layer in direct contact with human skin, effectively provides insulation from heat and cold [232]. Firefighter protective clothing is generally thick and heavy to insulate against heat; however, this characteristic may lead to discomfort due to a lack of breathability. Organic phase change materials (mPCM) in textiles regulate temperature by absorbing excess body heat and releasing it upon cooling, thereby maintaining an optimal comfort temperature of approximately 28 °C [233]. Phase Change Materials (PCMs), which consist of organic or inorganic compounds, are capable of absorbing and releasing substantial amounts of latent heat during the phase transitions between solid and liquid states [234]. The integration of phase change materials (PCM) into textiles presents a viable approach to improving fabric performance and thermal comfort under extreme temperature conditions. Fabrics incorporate phase change materials (PCMs) to prevent excessive heat transfer to the body in high-temperature conditions. In extremely cold conditions, phase change materials (PCMs) in textiles enhance thermal resistance, thereby preventing heat loss from the body to the surrounding environment [235,236]. Textile engineering has developed thermo-regulated smart fabrics that use nano-PCMs to autonomously adapt to environmental changes. Techniques such as coating, lamination, and fiber technology incorporate Nano-PCMs into textiles. Micro or nanocaps. We advise micro or nano encapsulating phase change materials (PCMs) to mitigate leakage during Sportswear, space suits, medical apparel, and lab coats utilize smart textiles [237, 238]. Fabrics incorporating nano PCMs exhibited enhanced stiffness and

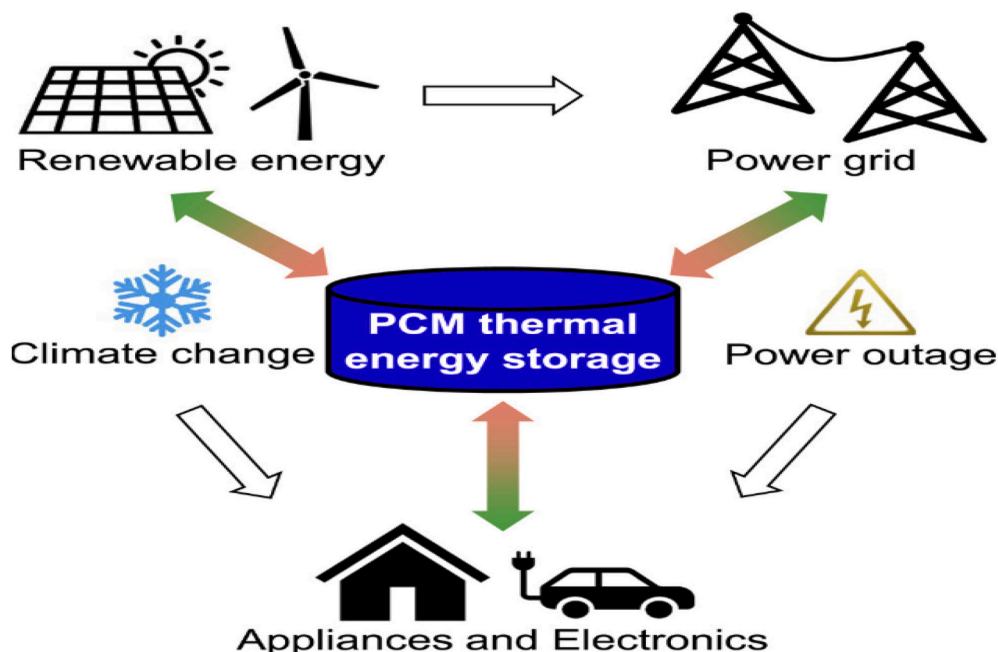


Fig. 10. PCM thermal energy storage [127].

roughness relative to those lacking nano PCMs; however, there was a slight decrease in resilience and smoothness. Researchers found that the physical and mechanical properties of VPWR fabrics incorporating nano PCMs were superior to those of fabrics lacking nano PCMs [231]. Fig. 11 shows the diverse applications of Phase Change Materials (PCMs) in textiles, including medical, sportswear, bedding, footwear, automotive, fire-retardant, and building materials for thermal regulation [239].

4.3.1. Smart textiles

Smart fabrics have emerged as a crucial element in wearable technology and protective systems [240]. We can classify smart textiles as passive, active, or ultra-smart, integrating mechanical, thermal, chemical, or hybrid technologies. These sophisticated textiles can be used in sectors like sportswear, space suits, low-temperature rescue equipment, automotive applications, medical devices, military gear, extreme

weather apparel, health and safety equipment, and sensor-based systems [241]. Smart textiles are advanced materials that can detect environmental or external stimuli and adjust accordingly while maintaining the properties of conventional fabrics. When these fabrics are mixed with phase change materials (PCMs), they become thermoregulated because they use the latent heat that is created and released during phase changes [240,242]. Phase-change materials (PCMs) that undergo transitions within a temperature range marginally above and below human skin temperatures are optimal for textile applications. The distinctive property of PCMs renders them appropriate for the development of protective textiles suitable for all seasons. Owing to the dynamic characteristics of phase change, these materials incessantly transition between states in response to environmental temperatures and the body's activity level, thereby maintaining ideal comfort [243–245]. Fig. 12 illustrates PCM-based smart fabrics that store solar heat during the day



Fig. 11. Various applications of PCMIC [239].



Fig. 12. Mechanism of smart textile [215].

(melting phase) and release it at night (freezing phase), helping maintain comfortable body temperatures [215]. Table 4 delineates the utilization of nano PCMs.

The washability and durability of NePCM-integrated textiles are influenced by several key factors, including the properties of the textile substrate, the integration method of the phase change materials (PCMs), and the specific characteristics of the PCMs themselves. The substrate's properties significantly affect the washing results, as different materials respond variably to washing processes, impacting the overall durability of the integrated textiles [153]. The integration method is crucial, as it determines how well the PCMs are embedded within the textile matrix, influencing both the mechanical stability and the thermal performance of the textiles during washing [174]. Additionally, the inherent properties of PCMs, such as their latent heat storage capacity and thermal stability, play a vital role in maintaining the functionality of the textiles post-wash. The lack of standardized wash testing methods for e-textiles,

Table 4
The application of nano-PCMs.

Sector	Application of Nano-PCMs
Building & Construction	Nano-PCMs are integrated into building materials (e.g., walls, roofs, and floors) for enhancing thermal energy storage, improving insulation, and stabilizing indoor temperatures, thus reducing heating and cooling loads [246–250].
Smart Windows & Facades	Nano-PCMs can be used in windows and facades to control solar heat gain and thermal transfer, improving energy efficiency by allowing dynamic thermal control based on external weather conditions [251–253].
Electronic Cooling	Nano-PCMs are employed in electronics to absorb and dissipate excess heat generated by devices, improving performance and preventing overheating in microchips, processors, and data centers [254–256].
Renewable Energy Storage	Nano-PCMs are incorporated into renewable energy systems (e.g., solar and wind) for storing excess energy in the form of latent heat, enabling efficient load balancing and ensuring a stable energy supply when demand fluctuates [257,258].
Solar Thermal Energy Storage	Nano-PCMs are used in solar thermal systems to store solar heat during the day and release it during cooler periods, increasing the overall efficiency of solar thermal energy systems and prolonging heat supply [259–261].
Smart Textile & Apparel	Nano-PCMs are applied in smart textiles for temperature regulation, where they absorb excess body heat and release it when the temperature drops, offering greater comfort and enhancing performance in extreme weather conditions [262,263].
Packaging & Transportation	Nano-PCMs are integrated into packaging materials to maintain desired temperature levels for sensitive products (e.g., food, pharmaceuticals), ensuring safe transportation by stabilizing temperature during transit [264–266].

including those integrated with PCMs, further complicates the assessment of washability, highlighting the need for standardized protocols to ensure consistent and reliable evaluation [175]. Moreover, common failure modes in e-textiles, such as increased resistance and mechanical degradation, are exacerbated by washing, necessitating targeted improvements in textile and circuit design to enhance washability [153, 174]. Overall, a comprehensive understanding of these factors is essential for developing NePCM-integrated textiles with improved washability and durability, ensuring their viability for various applications such as sportswear and medical textiles.

4.4. Electronics cooling

Due to their distinctive capacity to retain heat, phase change materials (PCM) have gained prominence in power electronics as an effective cooling medium [267]. It has served as a prevalent coolant in electronic cooling applications. [267,268]. Personal digital assistants (PDAs) and wearable computers are mobile devices that utilize heat storage units, specifically n-eicosane phase change material (PCM), for cooling purposes [128]. Other studies have established that a Thermal Energy Storage (TES) system can store energy during the solid-liquid phase transition [128,269]. This functionality renders TES systems applicable in many domains, such as solar HVAC and electronic cooling systems [128,129,270]. Another researcher discussed nano PCM. The attention of engineers in nano-PCMs has markedly increased due to their applications in thermal energy storage, cooling microelectronic systems, space heating in contemporary buildings, and smart fabrics for human comfort [122]. Fig. 13 outlines a solar irradiation setup with components like panels, a battery, and a heat exchanger, indicating a system for capturing and storing solar energy, possibly for heating fluids [163].

Applications for Nano-Enhanced Phase Change Materials (NEPCMs) include enhancing building heating and cooling, cooling electronic systems, refrigeration, and alternative thermal energy storage systems requiring rapid thermal cycling. Reports indicate that the application of nano-encapsulated phase change materials (PCMs) may reduce the energy demand for heating and cooling in residential buildings by around 1 % to 4 % [164]. It was the 1 % Al2O3 nanoparticle-infused Ne-PCM that could store the most heat, keeping the maximum input power as latent heat [271].

4.4.1. Advanced thermal management systems

To optimize the use of phase change materials (PCMs) in thermal storage and transmission, two primary thermal properties latent heat and thermal conductivity enhance the effectiveness of thermal management [264]. Mechanical and aeronautical engineering use a variety of materials, including phase change materials (PCM), for heat

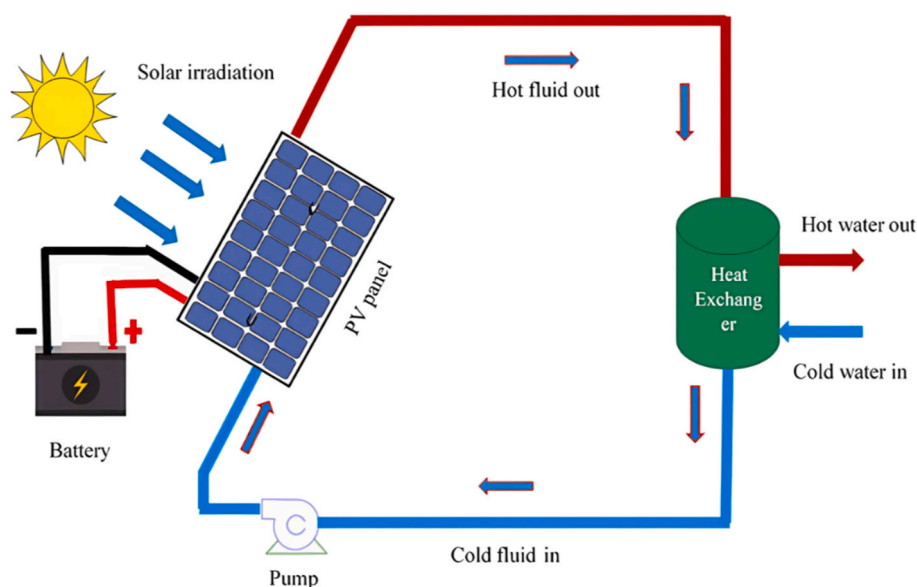


Fig. 13. Working principle of PVT system of electric cooling [163].

management and energy harvesting systems [246,260]. With the progression of nanotechnology, nanoconfined systems like nanofluids and nanocomposites offer novel perspectives on heat and mass transfer, which are beneficial for energy conversion and storage in thermal management applications. Phase change materials (PCMs) have a high latent heat of fusion, which means they can take in and give off large amounts of heat during phase transitions [254]. Fig. 14 lists equipment such as a vacuum gauge, thermostatic water bath, and data acquisition tools, suggesting a controlled environment for experiments, likely involving vacuum or pressure testing [270].

Researchers developed that there are many thermal management options including (a) active cooling which needs electricity and uses pumped water and (b) passive cooling which uses phase-change materials (PCMs) or geometric modifications (ex. Fins) [254]. Another system is Thermal Energy Storage (TES) which has the ability to improve the efficiency of many devices in different industries [256]. For

example, during the day, store energy in solar systems and release it at night. One technique for energy storage (ES) has been the usage of PCMs [252].

4.5. Packaging and transportation

The use of new technologies applications may enhance the function of food packaging materials and also used for maintaining of foods during transportation [248]. Additionally, by the use of PCM technology, cold chain energy management can be enhanced and reduce food temperature abuse [249]. Researcher showed that using nanotechnology of PCM helps to keep the food cold during transportation [250]. Another researcher suggested that salt hydrate PCM is used for packaging for transport and ice cream storage [253]. Nanoparticles can change the physical and mechanical characteristics of packaging polymers by enhancing their strength, durability, flexibility, barrier, and

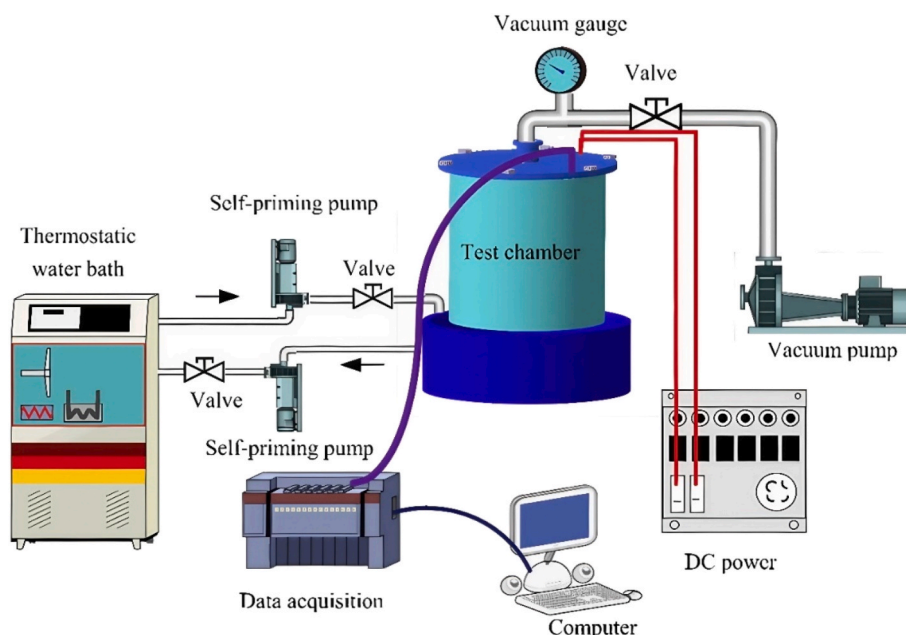


Fig. 14. Schematic diagram of the thermal conductivity test system [270].

reusing properties [265]. Fig. 15 contrasts conventional and nano based packaging, highlighting factors like UV exposure, microbial growth, and temperature/moisture effects, implying a study on material performance under different conditions [122].

Polymer-containing clay particles which are known as polymer nanomaterials offer improved properties for food packaging [266]. The use of nanotechnology of PCM that helps in the process of packaging in three different ways. That is improved packaging which has better physical and mechanical properties, active packaging which has antibacterial, antioxidative, and UV absorption properties) and smart packaging which has monitored/controlled food conditions) [262].

4.5.1. Temperature-controlled packaging

TCP means temperature-controlled packaging. Nano capsule powder was applied to show performance in managing the temperature inside a (TCP) box [258]. A technology that is known as Cold Thermal Energy Storage (CTES) which stores the cooling capacity below the normal temperature and is to be used at the time when cooling/cold demand is at its maximum [263,267].

Temperature-controlled packaging enhances user experience, reduces costs, and minimizes environmental impact. It helps to ensure Good Distribution Practice (GDP) standards, product quality protection, reduces losses, and ensures profitability [91]. Due to the PCM inability to maintain the desired payload temperature during manufacturing, the selected vendor for phase change material was unable to adapt the senior project's scope [272]. A synthesized nano capsule powder enhances the thermal buffering capacity of chocolate in temperature-controlled packaging which is used its effectiveness in managing temperature within (TCP) box [86]. The technology ensures high safety and efficiency in liquid-solid phase transitions so that they can effectively manage thermal management in the transportation of essential items. For example, food and vaccines [273]. Fig. 16 depicts a refrigeration setup with components like GFRP, VIP, cold-storage plates, and internal-circulation ducts, suggesting an energy-efficient cooling system for fresh-keeping or controlled-temperature storage [128].

The medium-sized insulated temperature-controlled container (ITCC), which measures 2 m in length, 1.8 m in width, and 1.72 m in height, employs construction and product-stacking techniques. Activating the fan draws cool air into the fresh-keeping compartment, which

transfers heat to the items, thereby elevating the air temperature and reducing the product temperature [128].

Researchers conducted extensive research and analysis on potential temperature control systems for drug shipping to clinics and researchers, concluding that the Credo Cube's technology was the most dependable and cost-effective option. The device incorporates vacuum-insulated paneling (VIP), which serves as a thermal resistor to minimize heat transfer from the environment to the payload [268].

4.6. Nucleation kinetics and interfacial effects

Nanoparticle interactions with phase change materials (PCMs) significantly influence nucleation kinetics and interfacial effects, which are critical for applications in energy storage and thermal management. The presence of nanoparticles can alter the nucleation phase by modifying thermodynamic parameters and enhancing or inhibiting nucleation rates, depending on their surface properties and composition [274, 275]. For instance, gold nanoparticles have been shown to enhance nucleation rates through electrostatic interactions, which lower the nucleation barrier, while the physicochemical characteristics of the nanoparticles and the substrate can further influence these processes [275]. Additionally, interfacial effects, such as the adsorption of molecules on the surface of PCMs, can either facilitate or hinder nucleation by affecting the desolvation energy barrier and the ordering of the liquid at the interface [276]. Overall, understanding these interactions is essential for optimizing the performance of nanocomposite PCMs in various applications.

4.7. Role of artificial intelligence (AI) for NePCMs

AI can significantly enhance the design and performance of nano-enhanced phase change materials (PCMs) for energy storage applications through various methodologies. Machine learning (ML) techniques, such as artificial neural networks (ANN) and adaptive neuro-fuzzy inference systems (ANFIS), have been employed to predict thermophysical properties and optimize synthesis conditions, thereby reducing costs and improving material characteristics [194,277]. For instance, ANN models have demonstrated high accuracy in predicting latent heat, achieving a correlation coefficient of 0.97 [195].

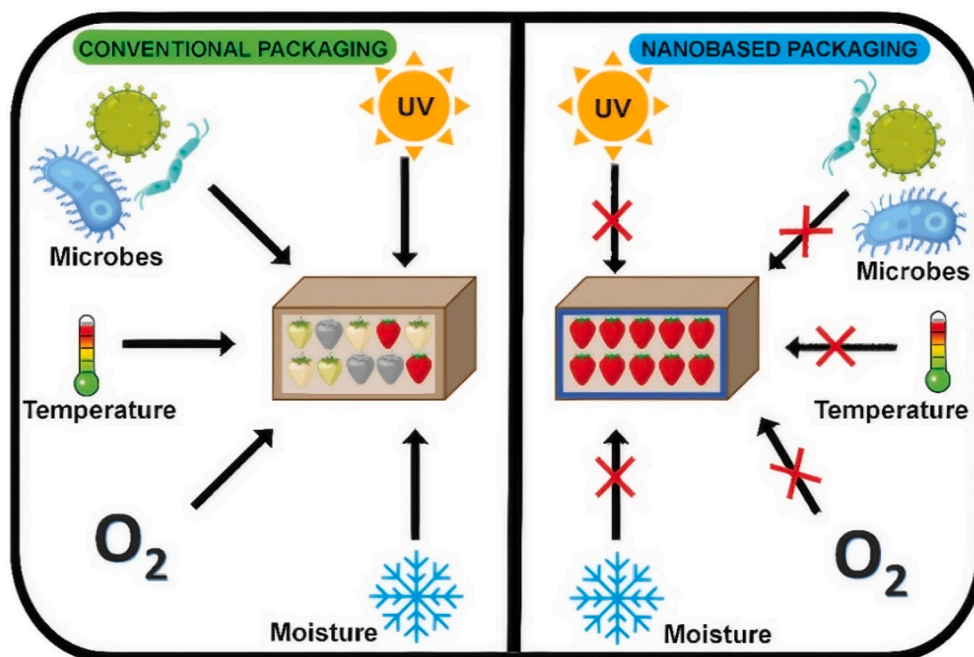


Fig. 15. Diagrammatic illustration of traditional food packaging and nano-based food packaging [122].

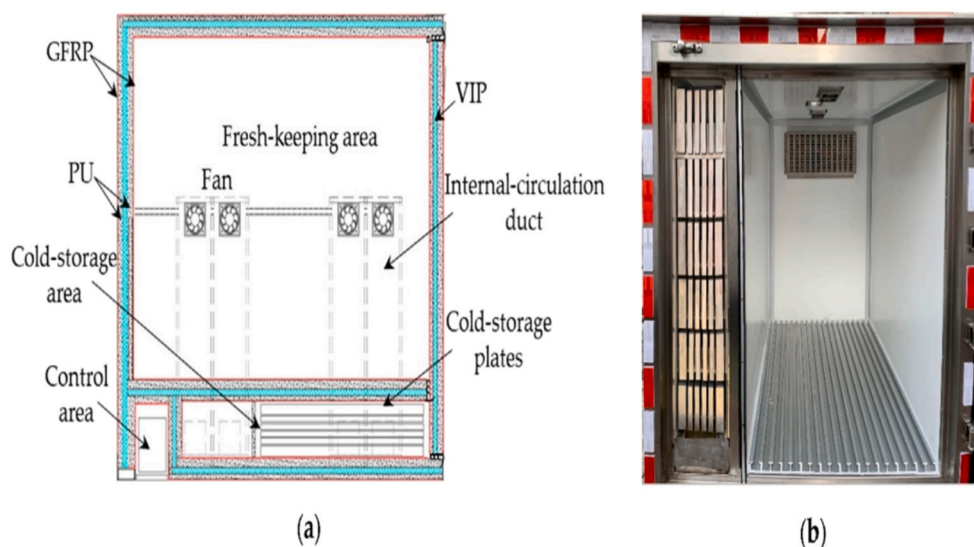


Fig. 16. Structure of insulated temperature-controlled container (ITCC): (a) vertical view and (b) side elevation [128].

Additionally, the integration of genetic algorithms with ANN has facilitated the identification of optimal configurations for heat pipes using nano-enhanced PCMs, resulting in improved thermal performance [222]. Furthermore, AI-driven optimization techniques, such as particle swarm optimization, have been utilized to maximize thermal energy and exergy outputs in photovoltaic-thermal systems incorporating nano-enhanced PCMs [340]. Overall, the synergy between AI and nanotechnology presents a transformative approach to advancing energy storage solutions, enhancing efficiency and sustainability [223].

Various studies demonstrate the effectiveness of ANN in forecasting key thermophysical properties such as latent heat, thermal conductivity, density, and dynamic viscosity [224]. Additionally, the use of targeted datasets in ANN models has led to exceptional accuracy in predicting thermal conductivity, with some models achieving over 99 % correlation with experimental data [278]. Overall, AI facilitates enhanced predictions of NEPCM properties, paving the way for improved thermal management solutions in various applications [243].

4.8. Next-generation eutectic hybrid PCM for thermoregulation

Next-generation eutectic, hybrid nanomaterial-based phase change materials (PCMs) have been integrated as a significant improvement to thermoregulation uses, since the limitations of traditional PCMs, namely low thermal conductivity and low storage capacity of energy have been addressed. Such hybrid PCMs, with nanoparticles such as graphene, carbon nanotubes as well as metal oxides, have higher thermal conductivities and energy efficiency and can be used in high-performance thermal management in electric and solar vehicles [279]. The systematic investigation of the improvement of phase change materials (PCMs) with hybrid nanomaterials (Ni-MWCNT dispersed nanocomposites) was conducted. Five different formulations were prepared and their thermophysical characteristics reviewed, which showed better energy-storage properties and complete chemical compatibility of the PCM and its host matrix. The emergence of MXenes and covalent organic frameworks (COFs) as new materials indicates that these materials will be extremely useful in energy-storage and conversion technologies [280]. They are desirable supercapacitors and batteries due to their extraordinary electrical conductivity, versatile surface chemistry, and surface space. However, commercialization of them remains challenged such as large-scale synthesis, stability and integration into device architecture.

It is the current work that has therefore highlighted the special qualities of these nanomaterials, current issues and proposed future

opportunities to future studies and advancements in the field [280].

As an example, when expanding graphite and nickel ferrite nanoparticles are used as part of eutectic PCMs, their thermal characteristics, such as thermal conductivity which has increased 253.72 per cent, and stability which has improved more than 500 times, have shown significant improvements thus they can be used in solar thermal energy and electronic cooling applications [281]. Equally, when lauric acid PCMs are supplemented with hybrid nanoparticles like hexagonal boron nitride and multi-wall carbon nanotubes, thermal conductivity increased by 52 percent and latent heat storage was also improved, and this is useful in thermoelectric generator applications or in cooling heat sinks [282]. Moreover, Ag₂O nanoparticles incorporated in eutectic PCMs have demonstrated 105 -fold increase in thermal conductivity and stability of the performance over 1000 cycles which implies that it can be used in medium- and low-temperature applications [283]. Generally, the evolution of PCMs using hybrid nanomaterials provides a breakthrough in improving thermal performance and reliability of energy storage platforms, and has notable vast potentials in the long-term sustainable energy solution [284,285].

5. Obstacles and prospective trajectories

5.1. Expenditure and scalability

The deployment of NePCM may pose a cost constraint in certain applications. While NePCMs provide long-term cost reductions via enhanced energy efficiency, their production and installation expenses exceed those of conventional materials. This may constitute a significant obstacle to adoption, particularly in cost-sensitive sectors [122]. A revolution is expected in the energy storage market due to the maturation and cost reduction of phase change materials (PCMs), decreasing from \$16 to \$3 per kilogram. The capital cost for thermal energy storage (TES) is comparatively lower, ranging from \$70 to \$200 per kWh, in contrast to electrochemical storage systems, such as batteries, which range from \$200 to \$300 per kWh. By implementing integrated sensible/latent heat thermal energy storage systems, material costs can be reduced to approximately \$15 per kWh, resulting in an energy efficiency of around 90 % [128]. Nanomaterials are quickly getting better, and this has led to a new way to combine very small, highly conductive particles like metals, metal oxides, and carbon into nano-enhanced phase change materials (nePCM). This makes micro-convection and thermal conductivity much better.

One can store thermal energy to transform a substance's internal

energy into perceptible heat, latent heat, chemical heat, or a combination of these [129]. A thermal energy storage system necessitates specific attributes, including efficient heat retention, substantial storage capacity per mass and volume, minimal heat loss, durability, longevity, and cost-effectiveness [129].

Presently, the majority of research on nano-enhanced phase change materials concentrates on thermal components and their applications, particularly in thermal energy storage construction. Studies indicate that the thermal conductivity of nano-reinforced PCM surpasses that of the base material [270]. This incorporates a sophisticated nanoPCM to mitigate adverse behaviors, such as cooling [270]. Numerous studies have explored ways to enhance the descriptive characteristics and diverse applications of NEPCM, as detailed in the literature review. Several publications have emphasized the importance of conducting an economic analysis of the NEPCM system to demonstrate its economic feasibility in a variety of applications [163]. We must balance the elevated expenses associated with the utilization of specialized nanoparticles against their advantages, as they directly impact the feasibility of large-scale NePCM manufacturing. The installation of tools and gear for NePCM preparation constitutes an additional expense that enhances overall performance. However, this is a one-time investment that requires regular maintenance and routine equipment servicing [122]. Fig. 17 contrasts passive (PCM, fins) and active (air, oil, water) thermal management methods in BICPV systems, highlighting different approaches to heat dissipation and energy efficiency [286].

Recent advancements in green nanotechnology have introduced innovative synthesis approaches that address production costs and market expansion barriers in the nanomaterials sector. Green synthesis methods, which utilize biological entities such as plants, bacteria, and fungi, offer environmentally friendly alternatives to traditional chemical processes, significantly reducing hazardous byproducts and associated costs [287]. These methods not only minimize environmental impact but also enhance the scalability of nanoparticle production, making it more economically viable for industrial applications [65]. Furthermore, the integration of green chemistry principles allows for the development of nanocomposites that are both cost-effective and sustainable, thereby expanding their market potential [287]. Despite the promising benefits, challenges remain in optimizing synthesis parameters and ensuring consistent quality at scale, necessitating ongoing research to fully realize the commercial viability of these green approaches [115].

The scalability and practical synthetic methodologies for nano-enhanced phase change materials (NePCMs) face several significant challenges. Key issues include high production costs, stability, and environmental concerns, which hinder widespread adoption in thermal energy storage and other applications [288]. Additionally, achieving precise control over nanoscale features while scaling up to macroscopic

structures remains a critical hurdle, necessitating advanced assembly techniques that integrate multiple length scales [289]. Optimizing synthesis parameters is crucial, as it can enhance thermal conductivity without compromising latent heat capacity, with recent studies demonstrating substantial improvements through hybrid optimization strategies [290]. Furthermore, the development of phase-engineered composites has shown promise in addressing low thermal conductivity and shape stability, enhancing thermal energy storage capabilities significantly [291]. Overall, ongoing research is essential to refine these methodologies and improve the feasibility of NePCMs in practical applications [292].

5.2. Environmental and health concerns

Environmental contamination is currently one of the most significant global challenges. Researchers are investigating innovative methods to address air, water, and soil contamination [164]. Examples of pollutants include particulate matter, heavy metals, pesticides, herbicides, and fertilizers; oil spills; poisonous gases; industrial pollutants; sewage; and organic compounds [271,293]. Environmental alterations can employ a variety of materials, providing numerous ways to achieve this goal. The absorption and removal of contaminants is challenging due to the intricate mixing of diverse chemicals, significant crosslinking, and poor reactivity. Recent research has concentrated on employing nanomaterials for the advancement of novel environmental modification technologies [294]. We need to restrict and regulate smog emissions due to their impact on climate change, the growth and decay of flora, and the health of humans and animals. According to the National Academy of Sciences, the emission of chlorofluoromethanes and chlorine compounds into the atmosphere enhances the absorption and emission of infrared radiation. Continuous removal of heat from the Earth will disrupt the planet's temperature and climate. We have established global regulations to reduce organic vapor pollution in the atmosphere. The US Environmental Protection Agency's ambient air quality guidelines stipulate that the maximum 3-h concentration of hydrocarbon content must not exceed $1.6 \times 10^{-4} \text{ kg/m}^3$ (0.24 ppm) for more than one year [295]. Furthermore, NePCMs have demonstrated their capacity to revolutionize waste heat recovery operations [165]. NePCM optimizes energy usage and minimizes waste by effectively capturing and storing waste heat from high-temperature industrial processes, thereby significantly contributing to energy savings and the reduction of greenhouse gas emissions, which is essential for identifying potential challenges and limitations [122].

It is crucial to meticulously analyses and resolve the issues related to this rapidly evolving technology, even though its purpose may change. Numerous studies have demonstrated the potential health implications

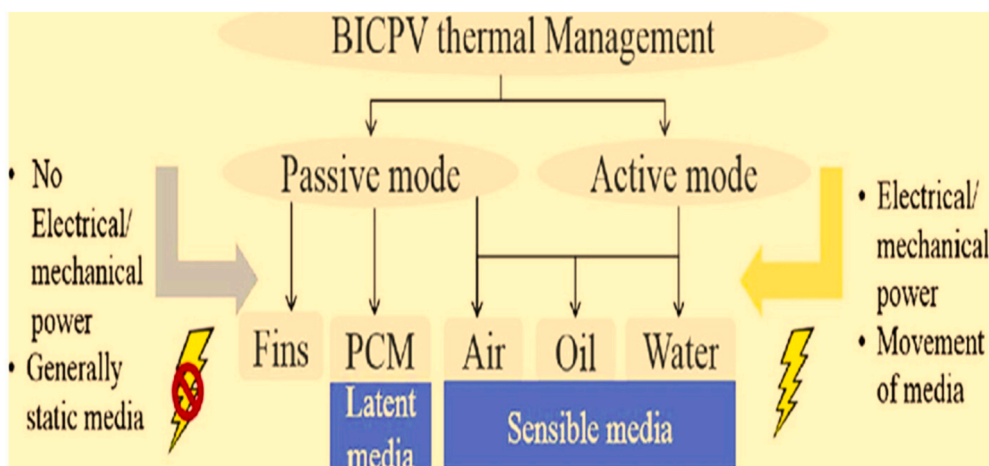


Fig. 17. BICPV thermal regulation options: Passive versus Active cooling mode with Latent and Sensible media [286].

of nanotechnology and underscored the necessity for prudence. One study examined the effects of nanoparticles on children's health, while others concentrated on the neurotoxicity of nanoparticles [296–298]. Moreover, the health implications of carbon nanoparticles and ultrafine particles necessitate additional research [299–301]. Despite its significant influence on numerous industries and society at large, public awareness of nanotechnology remains limited, particularly among high school students [302]. Consequently, it is imperative to inform individuals of the hazards and potential advantages of nanotechnology. Additional research is needed to fully understand why ultrafine particle exposure is harmful to children and adults in different ways. However, the unique physical properties of nanoparticles make them ideal for biological applications. Consequently, it is essential to explore the prospective advantages of nanoparticles in nanotechnology [303]. Fig. 18 shows heat transfer fluids (HTF) and PCM for thermal regulation, while the seventh introduces polymer therapeutics and metal nanoparticles, indicating applications in nanomedicine and targeted drug delivery [304].

The environmental benefits of using nanoparticle-based phase change materials (NePCMs) primarily stem from their enhanced thermal properties, which contribute to energy savings in various applications, particularly in building energy efficiency and photovoltaic systems. NePCMs, by incorporating nanoparticles like aluminum oxide, improve thermal conductivity, thereby facilitating more effective heat storage and release, which can lead to reduced energy consumption for heating and cooling in buildings [305,306]. For instance, studies indicate that NePCMs can significantly enhance the performance of photovoltaic panels, resulting in increased power output and improved energy efficiency [307]. Additionally, the integration of NePCMs in energy-saving glazing has shown potential for substantial annual heating energy savings, ranging from 5.1 to 41.7 kWh·m⁻²·year⁻¹, depending on climatic conditions [308]. However, the production impact of NePCMs must be considered, as the benefits of enhanced thermal performance may be offset by the environmental costs associated with nanoparticle synthesis and incorporation [305,309]. Thus, while NePCMs offer promising energy-saving capabilities, a comprehensive assessment of their lifecycle environmental impact is essential for sustainable application.

The current understanding of the toxicity, degradation, and environmental persistence of nanomaterials, particularly in the context of nanomaterials used in next-generation phase change materials (NePCMs), is critical due to their increasing prevalence and potential

ecological impacts [286]. Research indicates that engineered nanomaterials can accumulate in various environmental compartments, leading to toxic levels that impair ecological functions and human health, as many are nondegradable and persistent [304]. Despite a significant rise in ecotoxicological studies, gaps remain in understanding the mechanisms of toxicity, which include direct cellular damage, dissolution releasing toxic ions, and oxidative stress [310,311]. Furthermore, the life cycle of these materials poses challenges, as their environmental release and interactions with biota are not fully understood [312]. Thus, comprehensive risk assessments and the development of safer, more degradable nanomaterials are essential to mitigate potential environmental harm [313].

5.3. Integration and performance

When the temperature changes, phase change materials (PCM) keep working the same way [289]. Phase change materials (PCM) possess the ability for latent heat thermal energy storage (LHTES), serving as a novel method for thermal energy storage (TES) to address global energy demands. These materials undergo a phase transition, either melting or freezing, while conducting electricity and absorbing heat at a steady temperature [314]. Substances including water, paraffin, we regard substances such as water, paraffin, fatty acids, sugars, alcohols, and salt hydrates as effective phase change materials (PCMs) with diverse melting points suitable for low-temperature applications. y possesses a large latent heat capacity, absorbing and releasing thermal energy during the processes of melting and solidification. Organic phase change materials (PCMs) like fatty acids, paraffin, polyalcohols, sugar alcohols, and polyethylene glycol are suitable for low-temperature applications. Inorganic phase change materials (PCMs) such as platinum, nitrogen, sulphates, fluorides, and carbonates are applicable in high-temperature contexts. The chemical composition of pure phase change materials (PCMs), such as paraffin wax and fatty acids, makes them desirable for thermal energy storage (TES) because of their increased thermal storage density, cost-effectiveness, and renewability [315].

People frequently use inorganic shell materials such as silica, titania, calcium carbonate, polystyrene (PS), and zinc oxide. The literature suggests that people prefer inorganic shell materials over biological ones due to their advantageous qualities such as superior heat conductivity, robust tensile strength, chemical inertness, and nontoxicity [289]. The most straightforward approach involves the direct integration of PCM

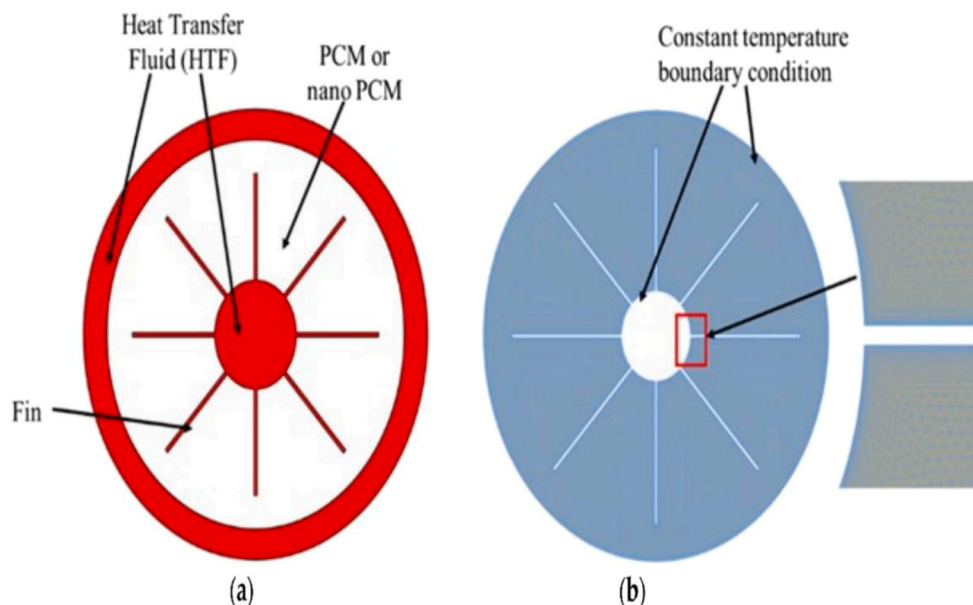


Fig. 18. Main characteristics of an ideal nano-enhanced phase change material [304].

into gypsum, concrete, or other porous substances to create a PCMBW-type composite. The volume of PCM within the pores is sufficiently small to inhibit the loosening of the solid PCM shell. The configuration of the porous material conducts heat to the pores [288, 310].

The implementation of phase change materials (PCMs) in the construction industry can yield substantial advantages for energy efficiency in buildings by modulating temperature under extreme weather conditions and decreasing power consumption, including the management of electrical loads [166]. The incorporation of phase change materials enhances the efficacy of energy storage systems and their associated heating and cooling applications. The primary application of PCMs in construction has occurred between 60 and 25°N and 40-25°S, emphasizing the enhancement of energy efficiency at the building level to improve interior comfort and decrease energy expenses [167]. Despite the diverse applications of PCMs, which are attributed to their dynamic characteristics and extensive temperature ranges, the residential sector holds the most significance, as buildings account for 45 % of global energy expenses. Energy usage in buildings fluctuates based on industrial and residential requirements. The need for power and heating has risen markedly in recent years. The demand for power has risen owing to the proliferation of solar energy [314].

5.4. Regulatory and standardization issues

Researchers study high-temperature phase change materials (PCMs) with melting points over 300 °C, which have the potential to store thermal energy in solar power plants. These comprise inorganic salts, eutectic salts, metals, and metal eutectics. Phase change materials (PCMs) with melting temperatures between 300 and 550 °C may serve as effective energy storage media for solar power plants utilizing high-temperature fluids (HTFs) such as oil, water/steam, and metal salts [290]. Organic phase change materials (PCMs) and inorganic phase change materials (PCMs) represent two prevalent categories of PCMs. Organic phase change materials (PCMs) have garnered significant interest owing to their favorable characteristics, including effective heat storage, minimal cooling during phase transitions, and low reactivity with most chemical reagents. Despite exhibiting high thermal conductivity.

The PCM mineral demonstrates favorable attributes such as latent heat storage capacity, cost-effectiveness, and non-flammability [292]. Conventional phase change materials (PCMs), encompassing both organic and inorganic types, exhibit low thermal conductivity, typically ranging from 0.1 to 0.6 W/m K. This level of conductivity is insufficient, resulting in slow charging and discharge rates. Various techniques can enhance the thermal conductivity of phase change materials (PCMs) by incorporating high conductivity materials [316]. Numerous solid-liquid phase change materials (PCMs) exhibit a phase change temperature range of 15–35 °C, making them appropriate for textile applications. These include polyethylene glycol (PEG), paraffins, hydrated inorganic salts, and their eutectic mixtures. Paraffins are the most suitable phase change materials (PCMs) for thermal management and temperature regulation in textiles, owing to their high latent heat and thermal conductivity. The force facilitates positive changes in alignment with human body comfort and is non-toxic. It is safe for human use, exhibits excellent chemical stability, and has low toxicity [291]. Table 5 provides a summary of regulatory and standardization concerns.

The safety regulations and guidelines for NePCM (Nano-enhanced Phase Change Materials) in textile items and building materials must prioritize flammability standards, health impacts of nanoparticles, and environmental safety. Textiles are inherently flammable, necessitating stringent flammability regulations to mitigate risks associated with accidental ignition, which can lead to fatalities in fires [141]. Additionally, the use of engineered nanoparticles in textiles raises concerns regarding their potential health effects and environmental impact, highlighting the need for comprehensive risk assessments and

Table 5

An overview of regulatory and standardization issues.

Key words	Explains
Cost and Scalability	Continued investment in R&D is crucial for overcoming scalability challenges and reducing production costs. Public and private funding can support innovative approaches to PCM development [317].
Environmental and health concerns	The long-term effects of exposure to nanomaterials are not fully understood, necessitating further research to assess potential chronic health impacts [318].
Integration and Performance	Nanotechnology can enhance the thermal conductivity of PCMs by incorporating nanomaterials, such as carbon nanotubes or graphene, which facilitate better heat transfer during phase transitions [319].
Regulatory and Standardizations	Nanotechnology refers to the manipulation of matter at the nanoscale (1–100 nm), which can significantly enhance the properties of materials, including PCMs used for thermal energy storage and management [312,313].

responsive regulations to ensure consumer safety [142]. Furthermore, addressing occupational exposure to microplastics during manufacturing is crucial, as it poses health risks to workers, necessitating protective measures and proper ventilation systems [152]. Collectively, these regulations should ensure that NePCM applications in textiles and building materials are safe for consumers and the environment.

5.5. Risks and regulatory

The use of nanotechnology in phase change materials (PCMs) presents several potential environmental risks, primarily due to the unique properties and behaviors of engineered nanomaterials (ENMs). These materials, such as carbon nanotubes and metal oxides, are increasingly utilized for their enhanced thermal properties in PCMs, but they pose significant environmental and health concerns. One major risk is the bioaccumulation of nanoparticles in living organisms, which can lead to toxicity and adverse effects on aquatic and soil ecosystems [320,321]. The release of nanomaterials into the environment can occur at various stages of the product life cycle, leading to the emission of nano-contaminants that may persist in the environment due to their stability and resistance to degradation [322]. Furthermore, the specific physical and chemical properties of these nanomaterials contribute to their potential toxicity, which is exacerbated by their high surface area and reactivity [322,323]. Despite the benefits of ENMs in environmental remediation and energy applications, their long-term environmental behaviors remain insufficiently understood, raising concerns about their persistence and the potential for respiratory hazards and toxicity to aquatic organisms [321,324]. The complexity of environmental conditions and the lack of comprehensive data on the fate and transport of nanomaterials further complicate risk assessments, highlighting the need for more effective research methodologies and the development of green nanotechnology solutions to mitigate these risks [323,324].

Regulatory frameworks significantly influence the development and commercialization of nanotechnology-based phase change materials by shaping the legal landscape within which these innovations operate. Current regulations often rely on pre-existing frameworks designed for conventional technologies, which may not adequately address the unique characteristics and risks associated with nanomaterials [325, 326]. The complexity of regulating nanoparticles stems from their novel physical properties, necessitating a nuanced approach that considers their lifecycle stages—from supply to end-of-life disposal [327]. Furthermore, the existing regulatory gaps, particularly at the international level, complicate the commercialization process, as stakeholders must navigate conflicting regulatory drivers and scientific uncertainties [328,329]. As such, there is a growing consensus that tailored regulatory

measures, including nano specific regulations for risk assessment and public disclosure, are essential to foster innovation while ensuring safety and environmental protection [325].

6. Conclusion and prospects

This study presents a comprehensive review and critical analysis of advancements in nano-enhanced phase change materials (NePCMs), emphasizing their transformative potential in thermal energy storage and temperature regulation across a broad spectrum of applications. The integration of nanotechnology into PCMs has addressed critical limitations such as low thermal conductivity, phase separation, and leakage. By incorporating various nanomaterials—such as carbon nanotubes (CNTs), graphene, metal oxides, and silica-based structures—NePCMs exhibit significantly improved thermal performance, stability, and energy storage capacity.

The research highlights the effectiveness of hybrid nanomaterials and nanoencapsulation techniques in enhancing the heat transfer properties, reducing supercooling, and improving long-term stability of PCMs. Applications in building construction, smart textiles, electronics cooling, renewable energy storage, and packaging systems demonstrate the versatility and real-world relevance of NePCMs. Moreover, this work recognizes the importance of addressing environmental and health concerns associated with nanomaterial usage, advocating for green synthesis approaches and lifecycle assessments to ensure sustainability.

An innovative contribution of this study is the proposal of artificial intelligence (AI) techniques, such as machine learning and neural networks, to predict NePCM properties and optimize synthesis and application processes. This cross-disciplinary integration opens up new opportunities for precise, cost-effective design and deployment of NePCMs.

Looking ahead, several key areas warrant further research. Firstly, scalable and economically viable production methods must be developed to transition from lab-scale formulations to industrial-scale applications. Secondly, long-term thermal cycling tests and environmental impact studies are essential to assess the durability and ecological safety of NePCMs in various operating conditions. Additionally, further exploration of hybrid nanomaterials and multi-functional composites can unlock new performance benchmarks.

The future of NePCMs will also benefit from the development of standardized testing protocols and regulatory frameworks to ensure safe integration into consumer and industrial systems. The application of AI and data-driven modeling tools will continue to be pivotal in accelerating material discovery and optimizing energy storage solutions.

In conclusion, this research underscores the significant promise of nanotechnology in revolutionizing phase change materials for next-generation energy-efficient and environmentally conscious applications. Continued interdisciplinary collaboration, supported by innovation in material science, data analytics, and sustainability frameworks, will be crucial in advancing the practical utility and global adoption of NePCMs.

CRediT authorship contribution statement

Md. Abdullah: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Writing – original draft, Writing – review & editing. **Mohammad Obayedullah:** Data curation, Software, Writing – original draft, Writing – review & editing. **Md. Abul Khair:** Data curation, Software, Writing – original draft, Writing – review & editing. **Md Habibullah:** Data curation, Writing – original draft, Writing – review & editing. **Md. Nahidul Islam:** Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Delowar Hossain:** Investigation, Visualization, Writing – original draft, Writing – review & editing. **Md. Shariful Islam Shuvo:** Formal analysis, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Further reading

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